The International Price of Remote Work*

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Abstract

We study how the price of remote work is determined in a globalized labor market using data from a large web-based job platform where workers worldwide compete for remote jobs. Remote work is frequently offshored from developed to developing countries, yet remote workers in rich countries earn higher wages. While wage gaps across countries persist in the remote market, they are much smaller than gaps in non-remote, local wages. This premium is not explained by workers' observable characteristics, occupations, or employers' locations. Instead, remote wages respond to exchange rate movements in the worker's own country, indicating they are partly determined by local labor market conditions. Using a quantitative model disciplined by our data, we study how shifts in global remote labor markets affect US wages across occupations and across remote and non-remote jobs. Eliminating foreign remote competition would raise US remote wages by up to 22.5% and non-remote wages by up to 10% in highly offshorable occupations.

Keywords: Remote Work, Offshoring, Wages, Exchange rates, PPP.
JEL Codes: F1, F2, F4, F6

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1 Introduction

An increasing number of jobs are being done remotely, a trend that accelerated dramatically during the COVID-19 pandemic (Bloom et al., 2022; Aksoy et al., 2022; Hansen et al., 2022). Remote work can be done from anywhere, which can make these jobs easier to offshore (Blinder and Krueger, 2013). By further integrating global labor markets, the rise of remote work can have a profound impact on wages across the world (Baldwin, 2016, 2019; ILO, 2021). Are wages equalized across remote workers located in different countries? How do these wages respond to international shocks? How are wages in exposed countries affected by foreign remote competition? Which remote jobs are more susceptible to offshoring? While these questions are central for understanding the future of wages in both developing and developed countries, there is limited research on how wages are determined in globalized labor markets.

This paper uses new data from a large web-based job platform to shed light on these questions. Web-based job platforms match employers and workers located around the world who trade tasks that are delivered remotely, providing a window into a globalized market for remote work. The number of such platforms has tripled over the past decade. By 2020, hundreds of web-based job platforms had facilitated millions of international transactions totaling over US$50 billion (ILO, 2021). The emergence of these platforms coincided with the dramatic growth in information and communications technology (ICT)-enabled service trade, which has quadrupled in the US since the year 2000 and now accounts for 70% (US$800 billion) of all US service trade.[1]

Our dataset is sourced from one of the largest platforms in the market today. It has several features that make it particularly well suited for our purposes. First, workers are located around the world and compete for the same jobs. These jobs can be done remotely, require little capital other than a computer, and encompass a wide range of occupations, ranging from accountants to web developers. This makes the platform the ideal marketplace for studying the international price of remote work. Second, the dataset is very rich: in addition to hourly wages, it contains extensive information on worker characteristics such as experience, earnings, quality ratings, and standardized test scores and certifications. This information is essential for understanding cross-country wage differences, as it facilitates the comparison of workers around the world. Third, the data record the workers' job histories on the platform (wages, earnings, and start date of each job), which are necessary for understanding how remote wages respond to shocks. Finally, the job histories contain

[1]US Bureau of Economic Analysis, Table 3.1, International Services (accessed September 30, 2021).

the employers' identities and locations, which, in conjunction with the workers' locations, allow us to identify which jobs are being offshored.

We begin by documenting three stylized facts to understand the degree of international integration in global remote labor markets. First, we show that labor demand on the platform is highly concentrated in high-income countries—most notably the United States—and labor supply is distributed globally and concentrated in lower-income countries. The majority of contracts are cross-border, suggesting that remote work is largely offshored from richer to poorer countries. We also show that the propensity to offshore varies substantially across remote occupations: translation, design, and web-related work are highly offshored, while writing and legal jobs tend to be sourced domestically.

Second, we document significant wage disparities across workers from different countries. For instance, the average wage of workers in India is approximately one-third of that of their US counterparts. In fact, the worker's country of residence accounts for at least a quarter of the total wage variance in the data. Furthermore, remote wages are strongly correlated with the GDP per capita in the worker's home country, with an estimated elasticity of 0.22. This elasticity is not accounted for by observable differences in worker and job characteristics, differences in the employers' locations, or the fact that workers work for different employers. To understand why wages can differ across workers from different countries, we field an original survey of US companies. This survey reveals that firms using online platforms treat workers from different countries as imperfect substitutes and employ them for distinct tasks. We show, however, that remote wages are more equalized across countries than non-remote wages, suggesting that cross-border remote work compresses—but does not eliminate—international wage gaps.

The preceding evidence suggests that remote workers operate in a shared global marketplace, yet earn wages that reflect the economic conditions of their home countries. This raises the question of how remote wages adjust to international shocks. For our third empirical finding, we study how remote wages respond to two types of shocks that arise naturally in global markets: exchange-rate movements in the worker's country and changes in the wages of foreign competitors.

We estimate exchange-rate pass-through (ERPT) into remote wages by relating changes in a worker's wage in US dollars to changes in the dollar exchange rate of the worker's country. We find that pass-through into dollar remote wages is low: the partial elasticity of dollar wages with respect to the exchange rate is about 0.20. This implies a high degree of pass-through into local-currency remote wages (roughly 80%). This is in sharp contrast to non-remote wages, which typically do not respond to exchange-rate movements at short horizons. This finding reinforces the notion that remote wages are determined in part by local labor market conditions: an increase in the dollar value of a worker's local wage, induced by exchange-rate movements, raises their dollar wage on the platform, but less than proportionally.

We then show that a worker's wage reacts to changes in the wages of other workers on the platform. We regress a worker's wage change on a sector-level wage index capturing changes in the wages of the worker's competitors, and address endogeneity concerns by exploiting the fact that workers in different sectors face competitors from different countries. In particular, we instrument for competitors' wage changes using variation in exchange-rate movements in competitors' countries. We find that workers adjust their wages in response to changes in competitors' wages with an elasticity of about 0.72. This implies that remote workers are exposed to shocks that affect their foreign competitors, and that these shocks transmit through this global labor market.

Taken together, these facts indicate that the global remote labor market is integrated, but imperfectly so. The market is sufficiently integrated that wages are equalized to an extent, creating earning opportunities for workers in developing countries while leaving workers exposed to global competition. At the same time, cross-country wage gaps persist that are not explained by differences in skills, occupations, or employer characteristics. Instead, remote wages remain partly anchored to local labor market conditions.

We interpret these patterns by developing a quantitative model where workers in each country choose whether to work locally or in a global remote market. In the model, remote wages can differ across countries because workers face different local outside options (local wages) and because workers from different countries are not perfect substitutes. The same forces imply that shifts in the dollar value of local outside options—such as those induced by exchange-rate movements—generate partial pass-through into dollar remote wages, and that shocks affecting one set of countries can transmit to others through competition in the remote market.

We then use the model to analyze how changes in the global remote labor market affect US wages. Following the methodology of Dekle et al. (2008), we express the model in gross proportional changes. This results in a system of equations that depends on exogenous shocks; the share of remote work in the wage bill; the share of remote work sourced from each country observed directly in our data; and structural supply and demand elasticities. We calibrate these elasticities to match the same reduced-form relationships we estimate in the data: the response of dollar wages to the exchange rate and the response of workers' wages to changes in competitors' wages.

We first quantify the impact of foreign remote competition on US wages by simulating an increase in home bias that reshores all foreign remote work to the US (which in our model is isomorphic to an increase in offshoring costs). The model predicts sizable gains for US remote workers—their wage gains range from 9% in Writing to 22.5% in Design-related occupations, while non-remote wages in these sectors increase between 1.7% and 10.3%. These gains are partially mitigated by a 0.83% increase in the price level. Intuitively, reshoring remote work increases the demand for US workers, bidding up their wages and drawing additional workers into remote employment (which expands by roughly 60% to 130%). We show that the US wage gains from reshoring vary widely with the offshoring exposure of workers' occupations. In an extended version of our model calibrated to more granular occupations, remote wage gains range from 2.8% in the least offshorable occupations to 45% in the most offshorable, with non-remote wages following a similar pattern.

Second, we simulate an expansion in India's remote labor supply resulting in a roughly 50% increase in the Indian remote wage bill. This exercise is motivated by ongoing large-scale digital infrastructure investments aimed at positioning India as a global digital hub. The expansion of India's remote labor supply exerts downward pressure on both remote and non-remote US wages. Because Indian workers are closer substitutes for US remote workers than for US non-remote workers, the decline in remote wages is significantly more pronounced, and is larger in occupations where Indian workers account for a larger share of the remote wage bill. This change in relative wages triggers a reallocation of US workers from remote to non-remote employment, reducing non-remote US wages by up to 0.7%.

Third, inspired by recent US programs aimed at expanding rural high-speed internet access, we analyze an expansion in US remote labor supply. The increased supply of US remote workers reduces remote wages but raises non-remote wages, as workers shift out of local labor markets. Interestingly, approximately half of the resulting expansion in the US remote wage bill comes from a reduction in the foreign remote wage bill rather than a reduction in the non-remote wage bill. This suggests that expanding domestic remote labor supply would serve to reshore work currently performed abroad. We also explore the consequences of a further expansion of these platforms, driven by a global increase in the demand for remote services.

Our paper relates to various strands of the literature. First, it is related to a rapidly growing literature that studies the rise of remote work and its consequences. Hansen et al. (2022) document a three-fold increase in vacancy postings for remote work between 2019 and 2022. Aksoy et al. (2022) use data from 27 countries to document work-from-home patterns around the world in 2022. Barrero et al. (2022) use survey data to estimate that remote work can moderate wage-growth pressures in the US by 2 percentage points over two years.[2] We contribute to this literature by documenting cross-country differences in wages across workers in a globalized market for remote work, and by using a quantitative model to evaluate the implications of shifts in remote labor markets.

Second, our paper is related to a large literature on how wages are affected by foreign competition, either through trade (e.g., Goldberg and Pavcnik 2007, Autor et al. 2013, 2016), offshoring (e.g., Feenstra and Hanson 2003, Hummels et al. 2014), or international migration (e.g., Borjas 2014, Card and Peri 2016). Blinder (2009) and Blinder and Krueger (2013) classify occupations according to their offshorability, and consider jobs that can be done remotely to be easily offshorable. Our paper lies at the intersection of these topics, as the cross-border contracts in our platform can be simultaneously interpreted as trade in services, offshoring, or "tele-migration". We show that in a globalized market for remote work, a worker's wage responds strongly to changes in the wages of foreign competitors. We also document substantial heterogeneity in the frequency at which remote work is offshored across occupations. Our quantitative model suggests that foreign remote competition has a significant impact on US wages in the most offshored occupations.

Third, we contribute to a large literature on international price and wage comparisons. The main source of international price comparisons is the Penn World Table (see Feenstra et al. 2015), while more recent papers make international price comparisons using online data (see, e.g., Cavallo et al. 2014, Gorodnichenko and Talavera 2017, and Cavallo et al. 2018). Data on international wages are more limited. Ashenfelter (2012) documents cross-country wage differentials for McDonald's employees. Hjort et al. (2019) document that multinationals' wages around the world are anchored to wage levels at headquarters, while Hjort et al. (2022) use a database covering compensation for 300,000 middle managers to show that their wages vary little across countries. Within the US, Hazell et al. (2022) show that large firms post similar wages across locations. We contribute to this literature by providing international wage comparisons for remote workers. We show that despite the global nature of this marketplace, there is pervasive dispersion in wages across observationally equivalent workers who are located in different countries.

[2] There is a separate literature that uses data from remote job platforms to study topical questions in labor economics. Horton (2017) and Barach and Horton (2021) use experimental data from a large platform to study how minimum wages and compensation histories affect labor market outcomes. Stanton and Thomas (2015) use data from oDesk (now Upwork) to show that outsourcing agencies that intermediate between workers and employers have emerged in that market, while Dube et al. (2020) use data from Amazon Mechanical Turk to study monopsony power.

Finally, our paper contributes to an extensive literature on exchange-rate pass-through (see Burstein and Gopinath 2015 and the papers cited therein). Gopinath et al. (2020) show that in most countries, goods export prices in dollars are stable, and local-currency export prices move with the dollar exchange rate. Due to data limitations, that literature has focused almost exclusively on exchange-rate pass-through into goods prices. Our paper is the first to study pass-through into the price of tradable services (remote jobs). We show that ERPT into dollar wages is low, so remote wages denominated in domestic currency move almost one-to-one with the dollar exchange rate. In this respect, the global market for remote workers behaves similarly to the global goods market.

The rest of the paper is organized as follows. Section 2 describes the data. Section 3 compares remote wages across countries and documents how remote wages respond to international shocks. Section 4 develops an equilibrium model of remote hiring that rationalizes these patterns through location-specific outside options and competition in global labor markets. Section 5 quantifies the model and uses it to evaluate counterfactual shocks to the demand and supply of remote labor, and the last section concludes.

2 Data

2.1 Platform description and data collection

Web-based job platforms match workers and employers across the world who sell and buy services that are delivered online. We obtained our data from one of the largest web-based job platforms in the market today. The platform supports a wide range of industries, from accounting to web development, and hosts millions of registered workers and employers globally. The platform's transaction volume was approximately $2.5 billion in 2020 and $4.1 billion in 2023.

Workers registering on the platform must create a professional profile and specify their desired hourly wage (the 'ask' wage), which is displayed to potential employers in US dollars (USD).[3] Employers can engage workers either by posting job listings, to which workers apply, or by directly searching for workers whose profiles match their needs. The platform manages all billing and payments, with jobs typically paid within two weeks of completion. The platform's revenue model is based on fees charged to both parties: a

[3] All contracts are denominated in US dollars. However, the platform offers clients the option to settle these USD-denominated invoices in their local currency in several non-US countries.

percentage of the worker's invoiced earnings and a percentage of all payments made by the client to the worker.

We built our dataset by collecting the publicly available profiles of 100,023 workers on the platform who possess a completed profile and have documented positive earnings and prior job experience.[4] We collected two distinct data snapshots: one in January 2019 and a subsequent one in November 2020. The profiles contain the worker's 'ask' hourly wage alongside the following information:

General information: The platform displays the name and location (country and city) of each worker.[5] It also reports the type of jobs or 'occupations' that each worker can perform. The occupations are selected from a predetermined list of 91 occupations and are self-reported when workers create their profiles. In addition, workers can specify their time availability and provide a brief written description of their skills and interests in their profiles. From each profile, we extract a worker's unique (anonymized) identifier along with their location, occupation, and time availability, removing all other personal information.

Skills and certifications: Workers can list several predetermined skills and take online examinations through the platform to certify their expertise in certain areas, such as 'English to Spanish Translation'. The platform offers more than 200 different tests. Employers can observe the tests each worker takes, along with the scores and rank percentiles among the platform's population. We use the results from these tests as our primary measure of skills, because they are standardized across all workers.

Experience and quality: In addition to the information provided by workers, the profiles also display data drawn from each worker's interactions with the platform. Specifically, the platform records each worker's total earnings and total number of jobs completed. Additionally, it displays the average response time for each worker and the percentage of contracted jobs they have successfully finished, referred to as the 'success rate'. Finally, the platform certifies experienced workers with 'Top Rated' status, which is

[4] Since creating a profile is easy and free of charge, a large fraction of profiles appear to be ghost accounts with no registered activity on the platform. By focusing on workers with past earnings and experience, we exclude such inactive profiles from the analysis.

[5] The platform routinely sends workers and clients verification requests asking for documents that verify their residence (e.g., bank statements, credit card statements, and utility bills). The submitted address must match the location information that workers and clients entered on the platform.

awarded to successful workers who meet minimum criteria, including a 90% success rate and at least $1,000 in earnings in the preceding year.

Job histories on the platform: For each job a worker has started, the platform reports the job description and the total payment. If the contract was hourly, the dataset includes the transacted hourly rate and the number of hours worked. It also records the start and, if applicable, the end date of each job. Due to the complexity of the data collection process, job history data were obtained for a subsample of approximately 75,000 workers. These time-series data are used in Section 3.3 to analyze the response of remote wages to international shocks.

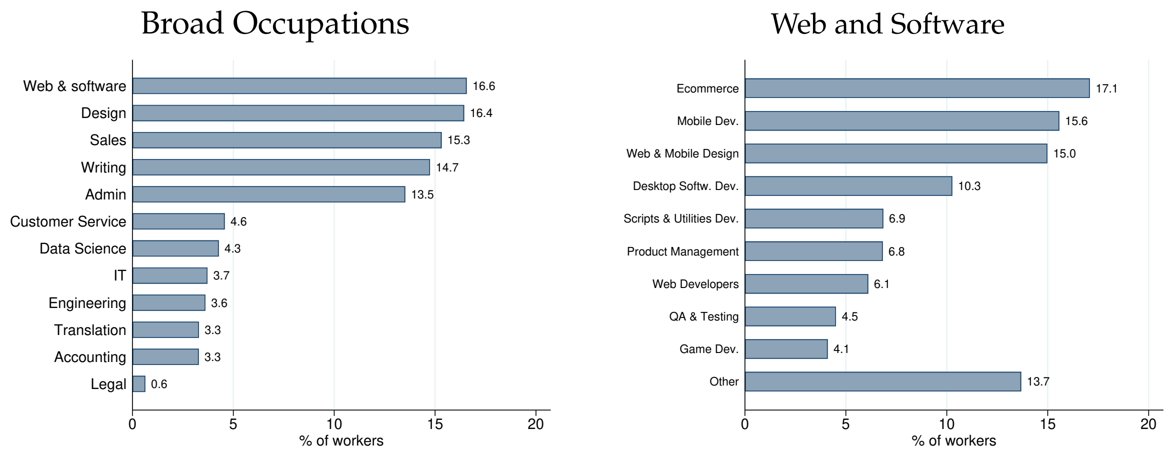
Employer locations: Job histories for a subsample of 22,152 of the 100,023 workers also include the geographic location of the employer. This information is used in Section 3.1, where we document the prevalence of cross-border job flows, and in Section 3.2, where we investigate the extent to which employers from different countries exhibit disparate wage-setting behaviors.

2.2 Summary statistics

The collected data encompass the profiles of over 100,000 workers located across 183 countries. Workers are classified into 12 broad occupational categories, which are further disaggregated into 91 detailed occupations (listed in Appendix Table A1). Figure 1 illustrates the distribution of workers across these broad occupations. The largest categories in the sample are 'Web and Software', 'Design', and 'Sales', accounting for 16.6%, 16.4%, and 15.3% of workers, respectively. In contrast, the 'Legal' broad occupation accounts for only 0.6% of the sample. Each broad occupation can be further disaggregated into detailed occupations. The right panel of Figure 1 provides an example of the disaggregation, showing that 17.1% of workers within the 'Web and Software' category are listed as 'E-commerce'.

Table 1 reports summary statistics for key variables. Worker 'ask' wages on the platform are high by international standards: the median and mean wages are $18 and $25 US dollars, respectively. There is, however, wide variation in wages: the gap between the 95th and 5th percentiles of the wage distribution is 2.8 times as large as the mean. The average worker in the data has completed 69 jobs and accumulated $18,667 US dollars in total earnings. Earnings are also highly dispersed, ranging from the 5th percentile of $20 to the

Figure 1: Workers by broad occupation



Notes: The left panel reports the share of the workers across the 12 broad occupations on the platform. The right panel reports the shares in each detailed occupation belonging to 'Web and Software'.

95th percentile of \$90,000. While these figures represent cumulative platform earnings, they are substantially greater (6–9 times) than the annual income per capita in countries such as India, Pakistan, or the Philippines, and are substantial even relative to US income per capita. This suggests that a large portion of the workers likely derive the majority of their income through the platform. Supporting this, 42% of workers report being available more than 30 hours per week, with an additional 33% available 'as needed'.

The platform allows workers to take standardized tests to signal their skills. The median (average) worker takes 3 (4) tests on the platform, and the standard deviation of the cross-test average score is 12% of the mean score. Finally, 41% of the workers in our sample are classified as 'Top Rated', and only 28% have a success rate of 100%.

Comparability of ask vs. transacted wages: As noted above, the dataset contains information on both the hourly 'ask' wage (listed on the worker's profile) and the hourly 'transacted' wage (recorded in the worker's job history for hourly jobs). A comparison between these two measures is presented in Appendix Figure A.1, which plots each worker's log ask wage (January 2019) against their average log transacted hourly wage (2018–2019). The figure shows that log transacted wages move nearly one-to-one with log ask wages: the slope of the relationship is 0.91. The intercept in the relationship is -0.02, which means that, on average, transacted wages are 2% lower than ask wages. While this marginal difference could stem from mechanisms like employer-worker bargaining during the hiring process, the overall quantitative relevance of such a divergence appears

Table 1: Summary statistics

	Mean	Median	St. Dev.	5 pct	95 pct
Ask hourly wage	25	18	27	5	75
Number of jobs	69	10	642	1	147
Total earnings	18,667	4,000	62,558	20	90,000
Number of tests	4	3	4	1	10
Average score	4.23	4.25	0.50	3.38	5

	Share of workers	Success rate	Share of workers
Top Rated	0.41	N/A	0.42
Agency	0.15	<70%	0.02
		[70%,80%)	0.03
Available as needed	0.33	[80%,90%)	0.07
Available < 30 hrs. per week	0.13	[90%,95%)	0.07
Available > 30 hrs. per week	0.42	[95%,100%)	0.11
Availability N/A	0.12	100%	0.28

Notes: The top of the table reports moments of the distribution of worker characteristics. Hourly wage refers to the ask wage specified in the worker's profile. Number of jobs and total earnings refer to a worker's cumulative experience up to January 2019. Number of tests and average score refer to the standardized tests offered by the platform to workers to certify their skills. The bottom of the table reports the share of workers classified as 'Top Rated' by the platform, the share of workers that belong to an agency, the distribution of the time availability reported by workers, and the distribution of success rates.

small.

Remote vs. non-remote wages for US workers: Finally, we compare remote wages on the platform to traditional non-remote wages for US workers across various occupations. The platform's occupations were manually matched to the US Standard Occupational Classification (SOC) categories using the corresponding descriptions (the concordance is detailed in Appendix Table A2). Data on traditional non-remote wages were sourced from the US Bureau of Labor Statistics (BLS).

Figure 2 compares hourly wages on the platform to those provided by the BLS for 38 SOC occupations represented in our data. Remote wages generally align with traditional wages for US workers, both ranging between $20 and $80 per hour, depending on the occupation. However, remote wages exhibit greater compression than traditional wages. The presence of a strong positive correlation between the two suggests that remote wages are, in part, determined by the earnings potential of workers in their local labor markets—a mechanism explored in detail in subsequent sections.

Figure 2: Remote vs. traditional wages for US workers

Notes: Each circle represents an occupation. The figure compares hourly average wages for US workers on the platform vs. wages in the BLS data for different SOC occupations. The estimated slope is 0.57 (0.10) and the R^2 is 0.43.

3 Characteristics of global remote labor markets

This section documents three characteristics of remote labor markets that are key for understanding their impact on global wages. First, remote jobs are frequently offshored, with the fraction of offshored jobs varies substantially by occupation. Second, while cross-country differences in remote wages are much smaller than those in non-remote wages, there are significant differences in remote wages among workers located in different countries, which are strongly correlated with the GDP per capita of the worker's home country. Third, remote wages in local currency are influenced by both exchange-rate fluctuations and changes in the wages of foreign competitors.

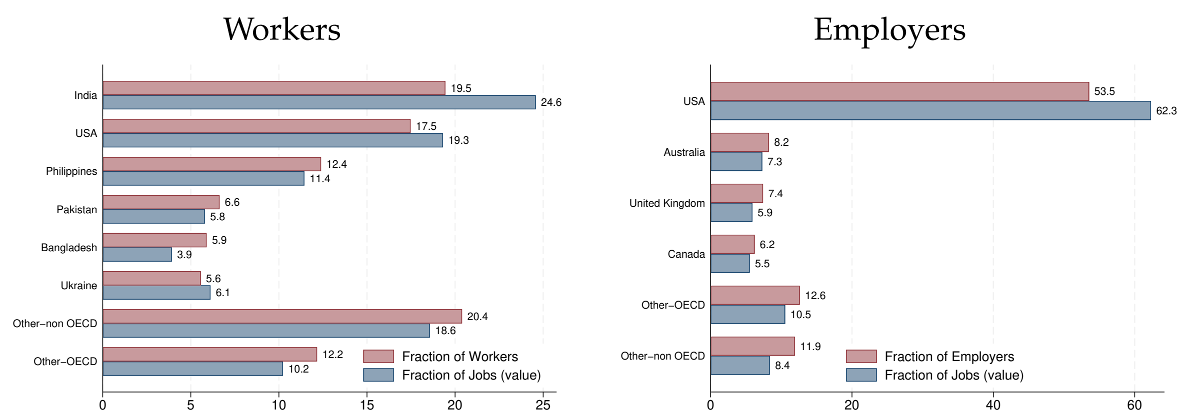
3.1 Prevalence of cross-border contracts

International flow of remote labor

We begin by examining the geographical origin of labor demand and supply in the remote market. The red bars in Figure 3 compare the geographical distribution of workers and employers in the data. Over 60% of workers are concentrated in five countries: India, the US, the Philippines, Pakistan, and Bangladesh. Labor demand is even more concentrated: 75% of employers are located in just four countries: the US (53.4%), Australia (8.2%), the UK (7.4%), and Canada (6.2%). The blue bars in the figure show the composition of remote labor supply and demand, measured by the value share of jobs, and paint a similar picture. In terms of value, US employers account for 62.3% of the jobs on the platform.[6]

While the US is a major source of both workers and employers, the flow of labor is predominantly offshored. Most employers (88%) are located in OECD countries, while most workers (70%) are located in non-OECD countries. Consequently, 86% of jobs (and 82% of the value of jobs) in our sample are offshored (i.e., the worker and employer are in different countries). Of these, 74% of jobs involve an employer from an OECD country and a worker from a non-OECD country.[7] Focusing on the US, 79% of jobs (and 74% of the value of jobs) generated by US employers are offshored. This highlights the extensive use of remote platforms to source labor from developing nations.

Figure 3: Distribution of jobs across workers' and employers' locations



Notes: The figure shows the distribution of jobs across workers' countries (left panel) and employers' countries (right panel).

Heterogeneity in offshoring propensity across occupations

Next, we analyze which types of remote jobs are most prone to be offshored. While existing offshorability measures rely on subjective judgments of how to classify the different attributes of a job (Blinder and Krueger 2013), we use the prevalence of cross-border contracts to obtain a direct empirical measure.

We use the US as our benchmark buyer location due to its status as the largest employer,

[6]These patterns are consistent with those in the Online Labor Index conducted by ILO. According to ILO, non-US workers in the main remote platforms are concentrated in India, Bangladesh, Pakistan, and the Philippines, and employers are mainly in the US and other OECD countries.

[7]In addition, 13% of jobs are among OECD countries, 10% among non-OECD countries, and 3% of jobs involve the worker located in an OECD country and the employer in a non-OECD country.

and define the offshoring share for an occupation j as the value share of US jobs performed by non-US workers:[8]

$$\mathcal{O}_j = \frac{\text{value of jobs in } j \text{ where cty. employer} = \text{US and cty. worker} \neq \text{US}}{\text{value of all jobs in } j \text{ where cty. employer} = \text{US}}. \tag{1}$$

The prevalence of cross-border contracts reveals substantial heterogeneity. Table 2, which reports $\mathcal{O}_j$ for the most and least frequently offshored occupations, shows that jobs for Technical Support Representatives are highly offshored ($\mathcal{O}_j = 0.93$), while jobs in Corporate Law are much less so ($\mathcal{O}_j = 0.24$). This heterogeneity suggests that a job's simple ability to be performed remotely is an insufficient proxy for its actual likelihood of being offshored. Appendix Table A3 reports how frequently jobs are offshored for the 91 occupations on the platform.[9]

Table 2: Most and least offshored occupations

Most offshored		Least offshored	
Technical Support Representatives	0.93	Corporate Law	0.24
ERP / CRM Specialists	0.92	Contract Law	0.29
Motion Graphics Freelancers	0.90	Grant Writers	0.29
Medical Translators	0.88	Academic Writers & Researchers	0.33
Legal Translators	0.86	Intellectual Property Law	0.33

Notes: The table reports the measure defined in equation (1) for the top 5 and bottom 5 occupations.

Fact 1 summarizes these findings:

Fact 1: *Remote jobs are predominantly offshored from developed (OECD) countries to developing (non-OECD) countries, and the propensity to offshore varies substantially across occupations. Labor demand is concentrated in the US, though labor supply is distributed across the globe.*

3.2 Cross-country differences in remote wages

We next document how remote wages vary across workers' and employers' locations. To do so, we estimate the following OLS regression using transacted wage data:

$$w_{fi} = \mathbb{C}_i + \mathbb{D}_f + \mathbb{I}_{i=f} + \boldsymbol{\beta}'\mathbf{X_i} + \varepsilon_{fi}, \tag{2}$$

[8] Appendix A.3 reports an alternative measure that captures the share of jobs that are offshored. The results are consistent across measures.

[9] Appendix Table A4 reports $\mathcal{O}_j$ for the Standard Occupational Classification (SOC) categories represented in our data.

where w_{fi} denotes the (log) wage paid by employer f to worker i in a given job. $\mathbb{C}_i$ and $\mathbb{D}_f$ are full sets of fixed effects for the workers' and the employers' countries, respectively. The omitted country category is the US, so these fixed effects measure the average wage earned by workers and paid by employers in each country relative to the US. $\mathbb{I}_{i=f}$ is an indicator for domestic contracts (worker and employer in the same country), and $\mathbf{X}_i$ is a vector of worker characteristics, including experience (log earnings, number of jobs), skills (number of tests, average score), quality ratings (Top Rated status, success rate dummies), availability, occupation dummies (all 91 detailed occupations), and agency indicators.

Worker and employer location

Figure 4a plots average wages across workers in each country relative to the US, obtained from the fixed effects $\mathbb{C}_i$ in equation (2), and the relative GDP per capita in each country with at least 100 workers with transacted wage data (see Appendix Table A5 for the estimates of $\boldsymbol{\beta}'$ in equation (2)). There is a very strong positive correlation between workers' remote wages and the GDP per capita of their home countries. The slope of this relationship is 0.22 (SE 0.03), and the R^2 is 0.58.[10] It is important to note that while pervasive, cross-country differences in remote wages are only about one-fifth the size of the differences in GDP per capita.

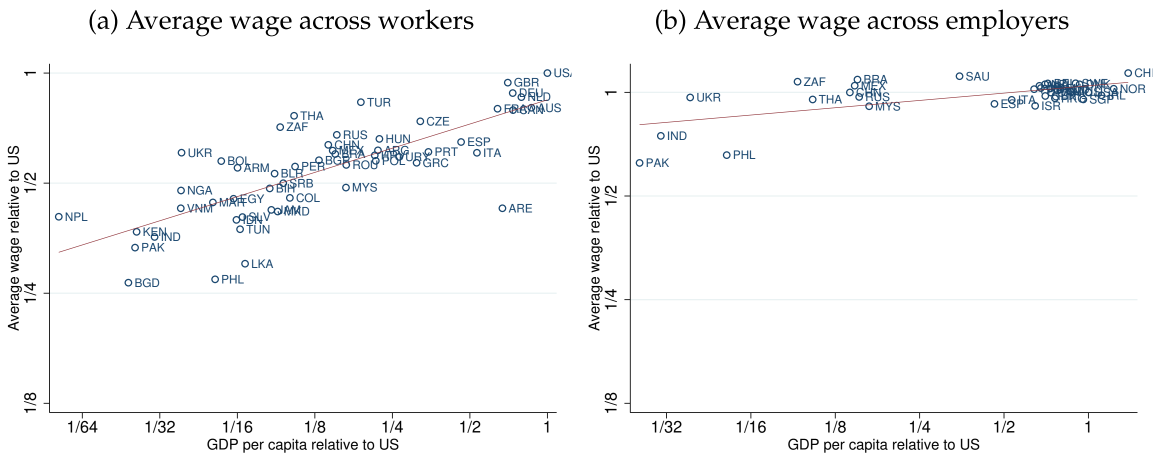
Figure 4b plots the average wages across employers in each country relative to the US, obtained from the fixed effects $\mathbb{D}_f$ in equation (2), for countries with at least 100 employers. The figure shows a very weak relationship between the remote wages paid by employers and the level of GDP per capita in their countries. This limited correlation is driven by a few outliers; only employers from Pakistan, India, and the Philippines appear to pay relatively lower wages than employers in the US.

A variance decomposition of equation (2) reveals that the workers' locations account for 31% of the total wage dispersion (calculated by summing its direct variance component and half of its covariance with the other covariates), exceeding the variance accounted for by all other controls in the equation combined. In contrast, the employers' locations account for only 0.5% of the wage variance, partly due to the high concentration of employers in a few countries.[11]

[10] Appendix Figure A.3 shows similar results using the larger sample of workers with available ask wage data, and Appendix Figure A.2 shows a similar relationship between non-residualized wages and GDP per capita.

[11] Appendix Table A6 reports the full variance decomposition. A regression of log wages on the set of country fixed effects $\mathbb{C}_i$ has an R^2 of 0.41, while the R^2 of estimating equation (2) with all the additional controls is 0.55.

Figure 4: Wages and GDP per capita relative to the US



Notes: The x-axes report the log of the relative GDP per capita in US dollars, taken from the World Development Indicators (WDI). Panel (a) and panel (b) plot $\mathbb{C}_i$ and $\mathbb{D}_f$ relative to the US obtained from the country fixed effects estimated in equation (2). The red lines show the linear fit to the data. The estimated slope is 0.22 (SE 0.03) in panel (a) and 0.07 (SE 0.02) in panel (b), and the R^2 values are 0.58 and 0.40, respectively.

External validity

To assess whether the cross-country wage gradient is specific to our platform rather than a general feature of global remote labor markets, we collected data on ask wages and locations for approximately 151,000 workers from one of the largest competing platforms. Estimating the gradient on the common sample of countries available on both platforms yields a slope of 0.14 (SE 0.02) on the competing platform, compared with 0.18 (SE 0.03) on our platform for the same set of countries and using ask wages without additional controls. The similarity of the two gradients indicates that the cross-country pattern documented in Figure 4a is a robust feature of remote labor markets and is unlikely to be the result of selection into our platform. We detail these results in Appendix A.4.1.

Disentangling sources of cross-country wage differences

The correlation between workers' wages and local GDP is persistent even after controlling for observable worker characteristics ($\mathbf{X}_i$) and the employer's country ($\mathbb{D}_f$). The correlation may reflect the impact of local income on remote wages, or the effect of unobserved factors (such as productivity differences) that correlate with local income, an issue we discuss in Section 3.3. Next, we perform several exercises to understand alternative potential drivers of this correlation. We detail these results in Appendix A.4.2.

Controlling for employer fixed effects: Estimating a version of equation (2) that includes unique employer fixed effects (for employers hiring more than one worker) shows that the strong relationship between (residual) remote wages and the GDP per capita of workers' home countries persists, although the slope of this relationship drops to 0.15 (SE 0.02). This confirms that even when working for the same employer, remote workers from richer countries earn higher wages.

Controlling for worker fixed effects (pricing to market): We evaluate whether workers price to market, that is, whether they sell their work at different prices depending on the employer's location. Estimating a version of equation (2) with worker fixed effects shows that workers are paid somewhat more when working for employers from richer countries, though the relationship is mild (slope of 0.05).

Trade costs: Incorporating controls for factors like geographical distance, time difference, common language, and legal origin into equation (2) does not significantly alter the primary results, suggesting that trade and transaction costs associated with these factors are not the main drivers of wage dispersion.

Border effects: We compare average wages across US states, residualized for worker characteristics, to their relative state GDP per capita. The pattern is similar to the cross-country findings: workers from richer states earn higher average wages (slope of 0.26, $R^2 = 0.48$). The similarity of the within-US pattern to the cross-country pattern suggests that the worker's location plays a large role in shaping wages, but not predominantly because of national border effects.

Comparison with non-remote wages and local prices: The relationship between remote wages and non-remote wages from similar occupations (sourced from the World Bank's International Comparison Program) resembles the pattern observed with GDP per capita. Furthermore, remote wages are higher for workers living in countries with higher local price levels.

Survey evidence on employer behavior: To directly investigate why employers are willing to pay different wages for remote workers from different countries, we fielded an original survey of 68 US-based companies that hire contract workers through online freelance platforms.[12] When asked why firms hire US workers despite higher costs, the dominant reasons cited were limited task-relevant English proficiency (53%), process complexity (33%), legal constraints (27%), and difficulty evaluating foreign credentials (25%). The survey also reveals how firms that *do* hire both US and non-US workers allocate tasks across workers. Open-text responses show a clear pattern: US workers are predominantly assigned compliance-sensitive tasks (e.g., legal consulting), client-facing roles (customer support, sales outreach), and work requiring cultural or market familiarity (copywriting for US audiences, HR compliance). Non-US workers handle standardized, back-office tasks (large-scale data entry, backend software testing, graphic design, social media management). These patterns are consistent with the notion that, even for remote occupations, workers from different countries have different comparative advantages across tasks.

We summarize these findings in Fact 2:

Fact 2: *Remote wages exhibit substantial cross-country differences, which are strongly correlated with the worker's home-country GDP per capita and persist after controlling for detailed worker and employer characteristics.*

3.3 Wage responses to international shocks

This section examines how remote wages are affected by international shocks. The preceding evidence suggests that remote workers operate in a shared global marketplace, yet earn wages that reflect features of their home countries. This raises a natural question: how do remote wages adjust when international conditions shift? To shed light on this, we exploit two shocks that arise organically in an international labor market setting: fluctuations in a worker's country-of-origin exchange rate and changes in the wages of foreign competitors.

Exchange-rate movements are particularly informative about the degree to which remote wages reflect local labor market conditions. Because exchange rates mechanically alter

[12] The survey was administered via Prolific in March 2026, targeting respondents with direct knowledge of their companies' use of major freelance platforms (e.g., Upwork, Freelancer.com, Fiverr, PeoplePerHour, or Guru). Of 92 respondents who passed an attention check, 68 (73.9%) confirmed platform use and constitute our analysis sample.

the dollar value of local wages—while local-market wages typically remain stable at short horizons—they shift the relative attractiveness of supplying labor to the platform. Analyzing how remote wages respond to this margin thus clarifies the link between remote work and the economic environments workers face at home. More broadly, the strength of these responses also helps interpret the degree of integration in this global labor market. In goods markets, a classic signal of integration is the sensitivity of prices to external shocks—particularly exchange-rate movements and foreign competitors' costs—as emphasized by Gopinath et al. (2011). Applying this logic to remote labor allows us to diagnose whether this emerging international market behaves more like a globally integrated tradable service market or a series of segmented local markets.

Sample construction for time-series analysis: As noted in Section 2, we focus on the sample of workers with rich job history data. We assign each job to one of four broad categories—'Admin and Sales,' 'Design,' 'Web and Programming,' and 'Writing'—using a machine-learning algorithm based on job descriptions.[13] We restrict our analysis to jobs with hourly wage.[14]

The start date of the job is reported at a monthly frequency, though a worker can start multiple jobs in the same month. We collapse the data at the monthly level so that the unit of observation is a worker-sector-month. After taking the difference between wages in two observations for the same worker within the same sector, this leaves a sample of 88,399 wage changes.

Finally, not all workers are observed in every month-sector combination, both because workers may not start new jobs in a sector in a particular month, and because our data contain only a subset of the jobs on the platform. With this caveat in mind, we denote the log-change in the dollar wage of worker i in sector j as $\Delta_s w_{ijt} \equiv w_{ijt} - w_{ijt-s}$ where the worker is observed in months t and $t-s$ (and not in between). More generally, we denote the s-period change in a variable by $\Delta_s x_t \equiv x_t - x_{t-s}$, and refer to the period itself as time-spell t_s. We summarize the distribution of wage changes in Appendix Table A7. In the following analysis, we use data on monthly exchange rate changes and CPI inflation obtained from the International Financial Statistics.

[13] The algorithm assigns a probability that a job belongs to each sector based on keywords from the job descriptions. For example, a job with the description 'looking for a grant writer' will likely be assigned to the sector 'writing' based on the keyword 'writer.' We detail the algorithm in Appendix A.2.

[14] About 50% of the jobs in this sample dataset are billed as 'fixed price' jobs, in which workers charge a predetermined price for completing a job. For these jobs, we observe how much workers are paid but not how many hours they work. We exclude these jobs from the analysis in this section.

Remote wages and exchange rate shocks

Following the international macroeconomics literature, we estimate partial exchange-rate pass-through (ERPT) elasticities using the regression:

$$\Delta_s w_{ijt} = \beta_1 \Delta_s e_{ct} + \beta_2 \pi_{ct_s} + \mathbb{C} \times \mathbb{J} \times s + \mathbb{T}_{jt_s} + \epsilon_{ijt_s}. \tag{3}$$

$\Delta_s e_{ct}$ is the change in the exchange rate in the worker's country, denominated in dollars per local currency. π_{ct_s} is the inflation rate in the worker's country. $\mathbb{C} \times \mathbb{J} \times s$ controls for country-sector-specific linear trends across different spell durations s. $\mathbb{T}_{jt_s}$ is a set of fixed effects that captures aggregate and sector-specific shifters for each time-spell duration $t \times s \times j$; and ϵ_{ijt_s} is the error term. Equation (3) is conceptually similar to the medium-run ERPT regressions estimated by Gopinath et al. (2010). The coefficients β_1 and β_2 are identified from both time and country variation in exchange rates and inflation. Note that, because the dependent variable is the change in wage for a given worker, any time-invariant component of worker quality drops out of the specification in first differences. The identifying variation comes from how the *same* worker's wage changes in response to exchange-rate and inflation movements.

Estimating (3) by OLS yields consistent estimates of β_1 if the error term ϵ_{ijt_s} is orthogonal to changes in exchange rates and inflation across countries. This requires changes in exchange rates to be uncorrelated with detrended sectoral productivity and supply and demand shifters at monthly frequencies. An extensive literature on the 'exchange rate disconnect' shows empirically that this restriction holds at short frequencies (see, e.g., Itskhoki and Mukhin, 2017).

We present the results from estimating equation (3) by OLS in Column 1 of Table 3. Standard errors are clustered at the sector-time-spell and country levels. The estimated partial pass-through elasticity is $\hat{\beta}_1 = 0.203$ and is statistically different from zero. This indicates that while dollar wages respond to changes in the dollar exchange rate, the elasticity is low. Correspondingly, remote wages expressed in local currency move substantially in tandem with the dollar exchange rate (with an implied elasticity of -0.797). This finding contrasts sharply with the notion that non-remote wages denominated in local currency do not respond to high-frequency movements in the exchange rate (see, e.g., Blanco et al., 2025).

From remote workers' perspective, a depreciation of their country's exchange rate represents a decline in the dollar value of their local-labor-market wages—assuming domestic-currency wages do not adjust rapidly. If high-frequency exchange-rate movements are uncorrelated with fundamentals such as productivity, the fact that remote wages respond to the exchange rate indicates that, even in a globally connected labor market, wages depend on the conditions workers face at home. Specifically, short-run changes in workers' outside options—induced by currency movements—translate into adjustments in the wages they receive on the platform. This pattern is consistent with remote work behaving like an internationally traded service whose price is partially disciplined by local labor market conditions. This interpretation is reinforced by the fact that the estimated $\hat{\beta}_1 = 0.203$ is similar in magnitude to the elasticities of remote wages with respect to proxies of local conditions (e.g., income per capita) documented in Section 3.2.

Remote wages and changes in foreign competitors' wages

We also study how wages are affected by changes in the wages of foreign competitors. To do this, we construct an index of the average wage changes in each sector, defined as the weighted average of mean wage changes within a sector-country pair, using market share as the weight: $\Delta_s w_{jt} \equiv \sum_c s_{jct} \mathbb{E}_c \left[\Delta_s w_{ijt}\right]$.

Obtaining such an index is challenging because the set of workers observed in our data changes over time. Thus, for any given time spell t_s, data on $\Delta_s w_{ijt}$ are not observed for many workers. We therefore approximate $\Delta_s w_{jt}$ as the change in the average wage observed across periods $t - s$ and t, after controlling for worker composition. More specifically, we first estimate:

$$w_{ijt} = \delta_{ij} + \delta_{jt} + v_{ijt},$$

where δ_{ij} and δ_{jt} are two sets of worker-sector and time-sector fixed effects, respectively. We then construct the series for the wage index as the series of the estimated time fixed effects: $\Delta_s w_{jt} = \Delta_s \delta_{jt}$.[15]

Once we have constructed a measure of sector-wide wage changes, we estimate the following equation.

$$\Delta_s w_{ijt} = \beta_1 \Delta_s e_{ct} + \beta_2 \pi_{ct_s} + \beta_3 \Delta_s w_{jt} + \mathbb{C} \times \mathbb{J} \times s + \mathbb{T}_{t_s} + \varepsilon_{ijt_s}. \tag{4}$$

Here, $\mathbb{T}_{t_s}$ denotes a set of fixed effects of each period by spell-duration combination ($t \times s$). The OLS estimates of (4) would be inconsistent if competitors' wage changes $\Delta_s w_{jt}$ were

[15]Through the lens of the model described below, we show that this procedure recovers, up to a first-order approximation, the time series of dw_{jt}.

correlated with the residual ε_{ijt_s}. We thus employ an Instrumental Variable approach. Exploiting the assumed exogeneity of the exchange rate, a natural instrument for $\Delta_s w_{jt}$ is the weighted average of exchange-rate changes across the countries that supply remote labor in sector j:

$$\Delta_s e_{jt} \equiv \sum_c s_{jct} \Delta_s e_{ct}, \tag{5}$$

where the identifying assumption is that $\Delta_s e_{jt}$ is uncorrelated with ε_{ijt_s}. We proxy the market share s_{jct} by the sample-wide share of jobs performed by workers from country c in sector j.

The instrument is relevant because exchange-rate movements shift the dollar value of workers' local outside options, prompting workers to adjust their remote wages. These individual wage adjustments aggregate into changes in the sector-wide competitor wage index $\Delta_s w_{jt}$. Crucially, different sectors have different country compositions of their remote workforce (e.g., the Web sector has a larger share of Indian workers than the Writing sector), so the same set of country-level exchange-rate movements generates different shocks to the competitor wage index across sectors. Appendix Figure A.8 documents that there is substantial variation in country composition s_{jct} across sectors, which is the source of cross-sectoral variation that drives the first stage.

The exclusion restriction is that, conditional on a worker's own exchange rate $\Delta_s e_{ct}$ and the other controls, the weighted average of exchange-rate changes in the worker's *competitors'* countries affects the worker's wage only through the competitor wage index. This rules out the possibility that exchange-rate movements in other countries directly affect a worker's productivity or the demand for that worker's specific services at short horizons—a restriction supported by the exchange-rate disconnect literature.

Table 3 Column 2 shows the results from estimating equation (4) by OLS, which controls for country-sector-specific linear trends and time-spell fixed effects but replaces the sector-time-spell fixed effects $\mathbb{T}_{jt_s}$ with the competitors' wage index $\Delta_s w_{jt}$. Standard errors are clustered at the sector-time-spell and country levels. The coefficients on the dollar exchange rate and inflation ($\hat{\beta}_1 = 0.212$ and $\hat{\beta}_2 = 0.197$) are very close to those in Column 1. The coefficient on the aggregate wage index is $\hat{\beta}_3 = 0.781$ and is statistically significant.

Column 3 reports the 2SLS estimates in which we use $\Delta_s e_{jt}$ to instrument for $\Delta_s w_{jt}$. The coefficients on the exchange rate and inflation remain almost identical to those in Column 2. The coefficient on $\Delta_s w_{jt}$ is 0.721 and is statistically significant at the 1% level. The bottom of Table 3 reports the F-statistic of the first stage, which is well above conventional critical values.[16] These results show that dollar wages do respond to changes in competitors' wages induced by foreign exchange-rate movements. In particular, the estimates imply that a 1% increase in the wages in country $c' \neq c$ increases wages in country c by $0.721 \times \left[s_{jc't} \times 1\%\right]$.[17]

We find that workers adjust their wages significantly when the wages of their foreign competitors change, implying that employers perceive workers from different countries as somewhat substitutable. The strong elasticity of own wages with respect to competitors' wages shows that shocks originating in one part of the world transmit quickly and meaningfully to workers elsewhere, revealing a high degree of integration in the remote labor market. Together, these results indicate that remote labor markets exhibit both sensitivity to local economic conditions and substantial cross-country competitive interactions—features whose implications will be analyzed through the lens of a model in the next section.

Table 3: Wage changes and international shocks

	(1)	(2)	(3)
	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$
$\Delta_s e_{ct}$	0.203***	0.212***	0.214***
	(0.058)	(0.052)	(0.053)
π_{c,t_s}	0.227*	0.197*	0.195*
	(0.120)	(0.103)	(0.104)
$\Delta_s w_{jt}$		0.781***	0.721***
		(0.073)	(0.256)
Observations	88399	88399	88399
Test $\beta_1 = \beta_2$	0.84	0.87	0.85
Specification	OLS	OLS	2SLS
F stat 1st stage			57.4

Notes: Column (1) reports the OLS estimates from equation (3), which contains period-by-spell-duration-by-sector fixed effects. Columns (2) and (3) report the OLS and 2SLS estimates from equation (4), respectively, and include period-by-spell-duration fixed effects. All columns include country-sector-specific linear trends across different spell durations s. The nominal exchange rate e_{ct} is measured in US$ per unit of local currency. Standard errors are clustered at the sector-time-spell and country levels. *: significant at the 10% level, **: significant at the 5% level, *** significant at the 1% level.

[16] Appendix Table A8 reports the first-stage regression in Column 1 and shows that the coefficient on $\Delta_s e_{jt}$ is statistically significant and contributes to the variation in $\Delta_s w_{jt}$.

[17] Table A9 in the Appendix reports the results obtained after imposing the constraint $\beta_1 = \beta_2$.

Robustness

This section summarizes several robustness exercises that complement the results presented above.

Conditioning on a wage change: The empirical analysis above assumes that workers' wages are flexible. To evaluate whether the response of dollar wages to the exchange rate is muted by nominal stickiness, we reproduce our analysis using the subsample of jobs with observed non-zero dollar wage changes. As shown in Column 3 in Appendix Table A9, the coefficient on the worker's own exchange rate increases from a baseline of 0.20 to 0.25. Although this indicates that wages are more responsive upon adjustment, the quantitative differences relative to our baseline estimates are small.

Alternative measures of competitors' wages: A potential source of concern is that the aggregate wage index $\Delta_s w_{jt}$ is by definition a function of each worker's wage change and may thus be correlated with the error term in equation (4). While this dependence vanishes asymptotically, it may introduce bias in finite samples. To address this potential endogeneity, we reestimate equation (4) using a leave-one-out index for competitors' wages, $\Delta_s w_{-ijt} \equiv \sum_{l \neq i} \frac{s_{ljt}}{1-s_{ijt}} \Delta_s w_{ljt}$, where s_{ijt} is the market share of worker i in sector j. Note that as individual market shares become negligible, $s_{ijt} \to 0$ (as they do in practice), $\Delta_s w_{-ijt} \to \Delta_s w_{jt}$. The results of this alternative specification, presented in Column 4 of Appendix Table A9, are nearly identical to our baseline estimates.

Placebo analysis: In our baseline estimates, we classify jobs into four broad sectors using the jobs' descriptions and a machine-learning algorithm, and assume that a worker's wage depends on the wages of other workers in the same sector. To validate this approach, we conduct a placebo analysis to determine whether wages respond to changes in the wages of remote workers from other sectors. We would expect workers to respond more strongly to competitors in their sector than to remote workers from different sectors. Using the classification probabilities generated by our algorithm, we identify the 'most distant' sector for each job—defined as the sector with the lowest estimated likelihood of membership. We then extend the estimating equation (4) to include the average wage change in the job's most distant sector as an additional regressor.

The results, reported in Column 5 of Appendix Table A9, demonstrate that the inclusion of this placebo variable has a negligible effect on the coefficient on same-sector competitors' wages. In contrast, the coefficient on the wage changes of the most distant competitors is much smaller in absolute value and is not statistically different from zero, as expected.

Alternative assumptions on country-specific trends: Columns 6 and 7 in Appendix Table A9 reestimate equation (4) using alternative controls for the country-specific trends. Column 6 does not control for country-sector-specific trends. Column 7 does not control for time-spell fixed effects. The table shows that our results are robust to the different ways we control for country-specific trends.

Estimation using the worker-level data: Finally, we reestimate partial ERPT elasticities using data on ask wages. As detailed in Section 2, these data have a more conventional panel format, where the wage posted by each worker is observed at two points in time: January 2019 and November 2020. The benefit of the ask wage data is that they allow us to mitigate concerns about selection, whereby exchange-rate changes affect the likelihood that a worker is hired for, or accepts, a job and therefore appears in our sample. Thus, we can estimate the partial pass-through elasticities from the following equation:

$$\Delta w_{ij} = b_1 \Delta e_c + b_2 \pi_c + \mathbb{S}_j + \mu_{ij}, \tag{6}$$

where Δx represents the change in a variable between the two periods, and $\mathbb{S}_j$ is a vector of sector fixed effects. We omit time subscripts to highlight that we only observe one wage change per worker. Here, the coefficients are identified from cross-country variation in exchange rates and inflation.

The results in Column 8 of Appendix Table A9 yield a pass-through coefficient of 0.084. This estimate is lower than that derived from the transacted wage data, further reinforcing our conclusion that pass-through into dollar wages is low. This occurs in part because ask wages are stickier than transacted wages. In fact, when we restrict the sample to observations with non-zero wage changes between the two periods—ensuring that worker accounts remained updated throughout—the exchange-rate pass-through elasticity increases to 0.14 (Column 9 of Appendix Table A9) and is strongly significant. This estimate is reassuringly close to our benchmark elasticity of 0.20–0.25 obtained from wage histories; crucially, the ask wage data cover a fixed set of workers over a fixed period, mitigating selection concerns.

Fact 3: *The elasticity of remote workers' dollar wages with respect to their countries' dollar ex-*

change rates is low. Consequently, remote wages expressed in local currency move with the exchange rate, with an implied elasticity of 0.80. *Furthermore, remote workers' wages respond positively to wage increases among their foreign competitors.*

4 Model

This section develops a quantitative model to study how remote wages are determined globally and how global remote competition affects US non-remote wages. The model combines (i) imperfect substitution across remote workers from different countries, as well as between remote and non-remote labor, and (ii) a supply of remote labor that depends on workers' local outside options. Since workers' earnings do not depend on the identity or location of the employer, we model the remote labor demand through a representative firm that recruits in a competitive labor market. We use the model to interpret the exchange-rate and competitor-wage elasticities estimated in Section 3 and to quantify the wage and welfare effects of counterfactual scenarios.

4.1 Conceptual framework

Preliminaries: We consider a world economy consisting of C countries indexed by c, with J sectors indexed by j and a numéraire outside good (money). One of the countries is the US, where a representative household consumes the outside good and goods from the J additional sectors. Each sectoral good is produced using US non-remote labor and a bundle of remote labor sourced globally, and the resulting goods are consumed only in the US. In each country-sector, a continuum of heterogeneous workers choose between supplying labor in the remote or the non-remote markets to maximize net income. We analyze how US wages and foreign remote wages are determined in equilibrium, taking non-remote, non-US wages as exogenous. Because the model is static, we omit time subscripts.

Preferences in the US: The representative US household has quasi-linear preferences over the sectoral goods (Y_j) and money (M, denominated in US dollars). Its utility maximization problem is:

$$\max_{M,Y_j} \quad M + \sum_j \bar{\gamma}_j \ln Y_j \tag{7}$$

subject to the budget constraint $M + \sum_j P_j Y_j \leq E$. Here, E denotes total dollar expenditures and P_j is the dollar price of good j. We assume that $E > \sum_j \bar{\gamma}_j$ so the solution is interior. Dollar expenditures (and revenues) in each sector are constant and determined by:

$$P_j Y_j = \bar{\gamma}_j. \tag{8}$$

Under these preference assumptions, sectoral expenditures are independent of the level of total expenditures E. Hence, in what follows, we do not need to specify how E is affected by changes in wages, trade imbalances, saving decisions, or other factors.[18] Our focus instead is on how sectoral revenues $\bar{\gamma}_j$ are allocated across US labor (remote and non-remote) and non-US remote labor.

Technologies: Sectoral output Y_j combines US non-remote labor and a bundle of remote labor sourced from multiple countries:

$$Y_j = \left[[1 - \bar{\alpha}_j]^{\frac{1}{\epsilon}} L_{jU}^{\frac{\epsilon-1}{\epsilon}} + \bar{\alpha}_j^{\frac{1}{\epsilon}} N_j^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}.$$

Here, L_{jU} denotes efficiency units of non-remote labor, which must be sourced from the US. In addition,

$$N_j = \left[\sum_c \left[A_{jc} N_{jc} \right]^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}},$$

is a CES bundle of remote labor from different countries, where N_{jc} denotes efficiency units of remote labor from country c and A_{jc} are productivity shifters. ϵ is the elasticity of substitution between remote and non-remote labor, and ρ is the elasticity of substitution between remote labor from different countries.

Let Ω_{jc} and B_{jc} denote remote and non-remote wages per efficiency unit paid to workers in country c and sector j. Cost minimization implies a demand for US non-remote labor in sector j given by

$$B_{jU} L_{jU} = [1 - \bar{\alpha}_j] \left[\frac{B_{jU}}{P_j} \right]^{1-\epsilon} \bar{\gamma}_j. \tag{9}$$

[18] Appendix A.6.1 extends the analysis to cases where changes in remote labor markets affect total expenditures on sectoral goods through their effect on P_j.

Similarly, demand for remote labor from country c in sector j is

$$\Omega_{jc} N_{jc} = \bar{\alpha}_j A_{jc}^{\rho-1} \left[\frac{\Omega_{jc}}{\Omega_j}\right]^{1-\rho} \left[\frac{\Omega_j}{P_j}\right]^{1-\epsilon} \bar{\gamma}_j, \tag{10}$$

where

$$\Omega_j \equiv \left[\sum_c \left[\Omega_{jc}/A_{jc}\right]^{1-\rho}\right]^{\frac{1}{1-\rho}}$$

is the price of the remote labor bundle in sector j.

Workers' decisions: In each country c and sector j, there is an exogenous measure M_{jc} of workers, each endowed with Z_{ij} efficiency units of labor. Workers choose whether to supply labor locally (non-remote) or remotely to maximize net income.[19] In the local labor market, workers earn a gross wage given by $Z_{ij} \times B_{jc}$, while in the remote market they earn $Z_{ij} \times \Omega_{jc}$. In turn, workers face idiosyncratic dollar costs of working remotely and non-remotely, respectively given by a fraction $\left[1-\kappa_{jc}\right]$ and $\left[1-\tau_{ij}\right]$ of their income.[20] A worker chooses to work remotely if and only if her net remote income exceeds her net income in the local labor market, i.e., $\left[Z_{ij} \times \Omega_{jc}\right] \kappa_{jc} > \left[Z_{ij} \times B_{jc}\right] \tau_{ij}$. Thus, there exists a cutoff

$$\bar{\tau}_{jc} \equiv \frac{\kappa_{jc}\Omega_{jc}}{B_{jc}}, \tag{11}$$

such that workers with τ_{ij} below this cutoff choose to work remotely. We assume that Z_{ij} and τ_{ij} are independently distributed and that τ_{ij} has cumulative distribution $F(\tau) = \tau^{\theta}$ with support $0 < \tau \leq 1$. The supply of remote labor from each country c in sector j is then given by

$$N_{jc} = M_{jc} \times Z_{jc} \times F(\bar{\tau}_{jc}) = \tilde{M}_{jc} \kappa_{jc}^{\theta} \left[\frac{\Omega_{jc}}{B_{jc}}\right]^{\theta}, \tag{12}$$

where $\tilde{M}_{jc} \equiv Z_{jc} M_{jc}$ and $Z_{jc} = \mathbb{E}\left[Z_{ij} | i \in c\right]$ are the total efficiency units of labor and the efficiency units of labor per worker in country c. Equation (12) implies a constant labor

[19] One can interpret net income as a source of expenditures E that can be used for consumption of the outside good by the representative household in each country.

[20] Given our assumption of quasi-linear preferences, this formulation is equivalent to assuming that workers face distinct utility costs of working remotely versus non-remotely.

supply elasticity given by θ. The supply of non-remote labor in each country-sector is given by

$$L_{jc} = \tilde{M}_{jc} - N_{jc}. \tag{13}$$

Equilibrium: A competitive equilibrium is defined as a set of prices and quantities, $\{\Omega_{jc}, N_{jc}\}_{j,c}$ and $\{B_{jU}, L_{jU}\}_j$ such that, given non-remote wages outside the US $\{B_{jc}\}_{j,c\neq U}$,

i. **Firms minimize costs:** Labor demand equations (9) and (10) hold.

ii. **Workers maximize net income:** Labor supply equations (12) and (13) hold.

iii. **Remote labor markets clear:** Demand (10) equals supply (12) in all countries c and sectors j.

iv. **US non-remote labor markets clear:** Demand (9) equals supply (13) in all sectors j in the US ($c = U$).

Characterizing equilibrium changes: We now show how the equilibrium changes in response to changes in foreign non-remote wages B_{jc}, remote productivities A_{jc}, and the distribution of costs for working remotely versus non-remotely (summarized by κ_{jc}). Let $\hat{X} \equiv X_1/X_0$ denote the ratio of variable X across two equilibria. Equilibrium changes are characterized by the following system:

$$\hat{\Omega}_{jc}^{\theta+\rho} = \left[\frac{\hat{B}_{jc}}{\hat{\kappa}_{jc}}\right]^{\theta} \hat{A}_{jc}^{\rho-1}\hat{\Omega}_{j}^{\rho-\epsilon}\hat{P}_{j}^{\epsilon-1}, \tag{14}$$

and

$$1 + \frac{N_{jU}}{L_{jU}}\left[1 - \left[\hat{\kappa}_{jU}\frac{\hat{\Omega}_{jU}}{\hat{B}_{jU}}\right]^{\theta}\right] = \hat{B}_{jU}^{-\epsilon}\hat{P}_{j}^{\epsilon-1}. \tag{15}$$

The corresponding changes in remote-labor unit cost and the sectoral price are given by

$$\hat{\Omega}_{j}^{1-\rho} = \sum_{c} s_{jc}\hat{\Omega}_{jc}^{1-\rho}\hat{A}_{jc}^{\rho-1}, \tag{16}$$

and

$$\hat{P}_{j}^{1-\epsilon} = \left[1 - \alpha_j\right]\hat{B}_{jU}^{1-\epsilon} + \alpha_j\hat{\Omega}_{j}^{1-\epsilon}, \tag{17}$$

where $s_{jc} \equiv \frac{\Omega_{jc} N_{jc}}{\Omega_j N_j}$ is country c's remote-wage-bill share in sector j, and $\alpha_j \equiv \frac{\Omega_j N_j}{\Omega_j N_j + B_{jU} L_{jU}}$ is the share of remote work in sector j's wage bill. Equations (14)–(17) jointly characterize changes in the equilibrium.

4.2 Comparative statics

This section analyzes the model's implications for cross-country dispersion in remote wages and the sensitivity of these wages to exchange-rate and competitor-wage shocks, and relates these implications to the empirical findings in Section 3.

Cross-country differences in remote wages

Using lowercase to denote variables in logs, let $w_{ij} \equiv \omega_{jc} + z_{ij}$ denote the log wage per unit of time of remote worker i in country c and sector j. The equilibrium log remote wage can be written as:

$$w_{ij} = \frac{\theta}{\rho + \theta} b_{jc} + \frac{\rho - \epsilon}{\rho + \theta} \omega_j + \frac{\epsilon - 1}{\rho + \theta} p_j + \varphi_{jc} + z_{ij}, \tag{18}$$

where $\varphi_{jc} \equiv \frac{1}{\rho+\theta} \left[[\rho - 1] \, a_{jc} - \tilde{m}_{jc} - \theta \ln \kappa_{jc} + \ln(\bar{\alpha}_j \bar{\gamma}_j) \right]$ collects exogenous country-sector-specific supply and demand shifters (productivity, market size, and remote work costs).

Equation (18) states that, within a sector, wage differences across workers can arise from differences in local wages b_{jc}, country-specific demand and supply shifters φ_{jc}, and workers' efficiency units z_{ij}. Note that if workers from different countries are perfect substitutes, $\rho \to \infty$, remote labor demand is perfectly elastic, and wage differences are driven only by productivity differences (z_{ij} and a_{jc}). If, instead, remote labor supply is perfectly elastic, $\theta \to \infty$, remote wage differences arise from differences in local wages b_{jc} and differences in z_{ij}. For finite values of ρ and θ, the elasticity of remote wages with respect to local wages is positive but less than unity, $\frac{\theta}{\rho+\theta} < 1$. Equation (18) underscores that, while our model is highly stylized, remote wages are tied to local labor market conditions insofar as both (i) the labor demand from individual countries is downward-sloping; and (ii) the labor supply from those countries is upward-sloping (see Moretti 2011 and Card et al. 2018 for a discussion of similar determinants of wage differences in the context of domestic local labor markets). Appendix A.6.2 provides alternative micro-foundations for such conditions.

The model rationalizes Fact 2 from Section 3.2. If local wages can be proxied by the country's GDP per capita, then equation (18) predicts a partial elasticity of remote wages with respect to GDP per capita of $\frac{\theta}{\rho+\theta}$.[21] However, the empirical relationship between remote wages and income per capita could also reflect other correlated local conditions (encapsulated in the shifters φ_{jc} and z_{ij}), if these other conditions are not fully captured in the controls in equation (2).

Wage responses to international shocks

To study responses to shocks, let dx denote the change in variable x. We decompose the change in non-US local wages denominated in units of the numéraire (USD) as:

$$db_{jc} = g_{jc} + \pi_c + de_c. \tag{19}$$

Here, g_{jc} is the growth of non-remote wages in constant local currency units, π_c is the local inflation rate, and de_c is the log-change in the exchange rate (denominated in dollars per unit of local currency). Equation (19) indicates that the dollar-denominated change in local wages is the sum of real wage growth, domestic inflation, and nominal currency fluctuations.

Let $dx_j \equiv \sum_c s_{jc} dx_{jc}$ denote the sector-j cross-country average change in a variable (for purely country-specific variables such as e_c or π_c, this collapses to the sector-j trade-weighted average). Differentiating equation (18) and combining with (19) yields:

$$dw_{ij} = \frac{\theta}{\rho+\theta} de_c + \frac{\rho-\epsilon}{\rho+\theta} dw_j + \frac{\epsilon-1}{\rho+\theta} dp_j + d\psi_{jc} + dz_{ij}. \tag{20}$$

with

$$dw_j \equiv \sum_c s_{jc} \mathbb{E}\left[dw_{ij} | i \in c\right] = \frac{\theta}{\theta+\epsilon} de_j + \frac{\epsilon-1}{\theta+\epsilon} dp_j + d\phi_j. \tag{21}$$

Here, dw_j is the sector-j index of wage changes in the remote market, while $d\psi_{jc}$ and $d\phi_j$ collect supply and demand shifters (see Appendix A.5 for a derivation). Equations (20) and (21) state that remote wages in units of the numéraire (USD) respond to exchange-rate movements with a partial elasticity of $\frac{\theta}{\rho+\theta}$. Intuitively, an appreciation of country c's currency raises country c's dollar value of local wages, which reduces country c's

[21] If local wages b_{jc} are correlated with local prices, the model also predicts that remote wages should be higher in more expensive locations.

supply of remote labor and raises equilibrium remote wages in that country and sector. Equivalently, the partial elasticity of wages in units of country c's currency is $\frac{\rho}{\rho+\theta}$. Furthermore, we can interpret the estimated effects of competitors' wages through the lens of our model. Holding equilibrium prices p_j fixed, the partial elasticity of country c sector j remote wages with respect to a change in country c''s wages is $\frac{\rho-\epsilon}{\rho+\theta} \times \frac{dw_j}{dw_{jc'}} = \frac{\rho-\epsilon}{\rho+\theta} \times s_{jc'}$, which is positive if the elasticity of substitution across workers from different locations exceeds the elasticity of substitution between remote and non-remote workers ($\rho > \epsilon$). The positive estimates presented in Section 3.3 suggest that this is the relevant case.

4.3 Calibration

To quantify how wages respond to shocks in our model, we require information about the initial share of remote workers in the wage bill in each sector α_j, the employment level of remote work relative to non-remote work in each sector $\frac{N_{jU}}{L_{jU}}$, the share of workers from each location c in the total remote wage bill in sector j, s_{jc}, and the share of expenditures in each sector $\gamma_j \equiv \frac{P_j Y_j}{E}$. We also need to assign values to the supply elasticity of remote work, θ, the elasticity of substitution between remote workers from different countries, ρ, and the elasticity of substitution between remote and non-remote workers, ϵ.

We implement our model in four broad sectors j that match those used in Section 3.3: Admin, Design, Web, and Writing. We obtain s_{jc} directly from our data reported in Section 3.1 as country c's share of the wage bill of US contracts in sector j. We use the May 2017 Current Population Survey (CPS) Contingent Worker Supplement to compute $\frac{N_{jU}}{L_{jU}}$ and (together with platform data) α_j, and γ_j.[22] This supplement asks CPS respondents about income earned through jobs performed via online platforms.[23] We proceed in three steps. First, we map each SOC 2010 occupation in the CPS to one of our sectors using the crosswalk in Appendix Table A10. Second, we classify workers as remote versus non-remote using the new platform-work questions in the supplement. For the outside sector, all workers are classified as non-remote. Third, we compute remote and non-remote wage bills using CPS population weights. Since the CPS only covers US workers, we scale up the US remote wage bill in each sector using the sector-specific share of US workers in the platform's total remote wage bill, computed directly from our data.[24] We compute α_j as

[22]Since remote work has become more common since 2017, this calibration likely understates the impact of our counterfactuals.

[23]Workers using online platforms to perform non-remote tasks (e.g., Uber drivers) are not considered remote workers.

[24]Specifically, we obtain the total remote wage bill as $\Omega_j N_j = \Omega_{jU} N_{jU} / s_{jU}$.

the adjusted remote wage bill divided by the total adjusted sectoral wage bill, γ_j as each sectoral adjusted wage bill divided by the total adjusted wage bill across all sectors in the CPS, and $\frac{N_{jU}}{L_{jU}}$ as the ratio of US remote to non-remote employment within the sector.

Elasticities: We set the elasticity of substitution between remote and non-remote workers $\epsilon = 3.6$ based on Davis et al. (2024). Appendix Table A12 presents a sensitivity analysis with respect to changes in the value of this elasticity.

We obtain the key elasticities θ and ρ by structurally interpreting the empirical results from Section 3 through the lens of our model. First, note that under the assumption that exchange rate changes are disconnected from fundamentals (productivities of remote workers) at short horizons, the results of Section 3.3 indicate that the partial ERPT elasticity of dollar-denominated remote wages is $\hat{\beta}_1 = 0.21$, which corresponds to $\frac{\theta}{\rho+\theta}$ in the model. Second, as noted above, the partial-equilibrium elasticity of remote wages with respect to a change in competitors' wages is $\frac{\rho-\epsilon}{\rho+\theta}$. However, our empirical results in Section 3.3 do not control for the general equilibrium effects of remote wage changes on p_j (they do control for the GE effect on p, since the regression specification includes time fixed effects). Thus, we follow an indirect inference approach to discipline ρ and θ. In particular, we simulate changes in exchange rates and map them into changes in non-US local wages according to equation (19), holding all other exogenous variables in our model fixed. We then estimate equations (4) and (5) using model-generated data and choose θ (together with the restriction that $\frac{\theta}{\rho+\theta} = 0.21$) so that the simulated elasticity of remote wages with respect to a change in competitors' wages matches its empirical counterpart of $\hat{\beta}_3 = 0.72$. This procedure yields values of $\rho = 24.6$ and $\theta = 6.7$. Such a high value of ρ implies that remote workers from different countries are very close substitutes within a sector, consistent with the strong estimated response to competitors' wage changes. We summarize our parameterization in Table 4.

5 Quantitative analysis

This section employs our calibrated model to quantify the equilibrium impact of several shifts in the global remote labor market. First, we isolate how foreign remote competition affects US wages by simulating a complete reshoring of all foreign remote work to the United States. Second, we evaluate a global increase in remote labor productivity to assess how the continuing expansion of remote work affects both remote and non-remote wages. Third, we simulate an expansion in India's remote labor supply to gauge the

Table 4: Parameterization

Elasticities	Value	Source
ϵ: Elasticity of substitution between remote and non-remote labor	3.6	Davis et al. (2024)
θ: Supply elasticity of remote work	6.7	Own estimates from Section 3.3
ρ: Elasticity of substitution between remote workers from different countries	24.6	Own estimates from Section 3.3

Initial shares (in %)	Admin	Design	Web	Writing	Source
s_{jc}: Share of workers from location c in sector j remote wage bill	30.0	28.2	22.0	43.5	Platform
α_j: Share of remote workers in the sectoral wage bill	2.6	13.4	8.1	3.4	CPS & Platform
γ_j: Share of expenditures in sector j in total expenditures	12.80	1.27	3.93	2.41	CPS & Platform
N_{jU}/L_{jU}: Ratio of US remote to non-remote workers in sector j	0.9	5.0	3.1	1.3	CPS

Notes: This table shows the calibration of the model. The first panel contains supply and demand elasticities. The second panel reports a subset of the shares required to conduct the counterfactual analysis based on hat algebra. Values for s_{jc} in the second panel are reported for $c =$ US.

impact of rapid growth in a major supplier country. This exercise is motivated by large-scale digital infrastructure investments aimed at positioning India as a global digital hub. Finally, inspired by recent US programs aimed at expanding rural high-speed internet access, we analyze an expansion in the US remote labor supply to determine whether such policies lead firms to reshore jobs currently held by foreign remote workers or, conversely, displace existing US non-remote workers.

5.1 Reshoring all remote jobs

We first examine the extent to which exposure to competition from foreign remote workers depresses US wages. To do so, we set the remote productivity of foreign countries to $A_{jc} = 0$ for all countries $c \neq$ US, effectively shutting down imports of remote labor. These parameters govern the degree of home bias in our model; this is isomorphic to imposing prohibitively high iceberg costs on remote imports.

The results, presented in the first panel of Table 5, show a significant impact on the US labor market. US remote wages increase substantially, ranging from 9.2% in Writing to 22.5% in Design, while US remote labor expands by roughly 60–130%. Note that these increases are larger in sectors with greater initial offshoring exposure (i.e., the share of the remote wage bill that is offshored, $1 - s_{jU}$, Table 4). US non-remote wages also rise significantly, between 1.7% and 10.3%, with larger gains in sectors with the largest share of foreign remote workers in the wage bill (computed as $(1 - s_{jU}) \times \alpha_j$ from Table 4). Taken together, average US wages across both remote and non-remote workers in Admin, Design, Web, and Writing increase by 1.9%, 10.4%, 7.7%, and 1.8%, respectively, indicating that foreign competition is substantially depressing US wages in these sectors.

Finally, eliminating foreign remote labor raises sectoral prices. The implied dollar cost of this price increase for US consumers (i.e., the compensating variation) is equivalent to 0.83% of their initial expenditures.[25] Consequently, the wage gains for both remote and non-remote workers in these sectors exceed the welfare loss of higher prices.[26]

5.2 Increase in the demand for remote labor

This exercise evaluates an expansion in the market for remote work driven by an increase in remote labor demand. In particular, we increase the remote-labor productivity parameter A_{jc} by a common factor across all countries and sectors to double the equilibrium wage bill of the remote labor market.[27] As a reference, as noted in Section 2, the transaction volume of the platform grew by 65% between 2020 and 2023.

The second panel in Table 5 shows that, given our high estimate of the supply elasticity θ, the expansion in remote sales is primarily driven by an increase in the quantity rather than the price of remote labor. Specifically, remote employment roughly doubles in all sectors, while remote wages rise by 4.7–8.9%. The cost of the remote labor bundle (not reported in the table) falls by roughly 27% across all sectors. The increase in remote work productivity has two opposing effects on US non-remote wages: first, the fall in remote costs reduces non-remote labor demand; second, the rise in remote wages pulls workers into remote work and reduces non-remote labor supply. In our calibration, the first effect dominates, resulting in a 2.0–6.6% decline in US non-remote wages. Finally, the prices of the sectoral goods decline, generating a welfare gain for consumers equivalent to 0.91% of their initial expenditures.

5.3 Increase in the supply of remote labor from India

We next examine the impact of an increase in the supply of Indian remote workers. This exercise is motivated by the Indian government's forecasts of a 48% growth in the digital

[25]The compensating variation CV satisfies $V(\mathbf{p}_1, E + CV) = V(\mathbf{p}_0, E)$, where $\mathbf{p}_0$ and $\mathbf{p}_1$ are the initial and final price vectors, and V(.) is the indirect utility associated with (7). Substituting the demand (8) in (7) we can solve for $CV = \sum_j \gamma_j \ln \hat{P}_j$. Under quasilinear preferences, the compensating and equivalent variations both equal the change in consumer surplus resulting from the price change. A positive CV indicates that consumers must increase expenditure to maintain their pre-change utility level, implying that the price change constitutes a welfare loss.

[26]The price increase still reduces the welfare of US consumers who do not earn income from these sectors.

[27]As shown in equations (9) and (10), A_{jc} shapes the firm's relative demand for remote versus non-remote labor.

economy between 2025 and 2030, driven by strategic investments in digital connectivity.[28] We simulate this scenario by proportionally reducing the cost of remote work in India across all sectors—specifically by increasing the parameter κ_{jc} for c = India—to target a 48% increase in the Indian remote wage bill.

The third panel in Table 5 presents the results of this exercise. The increased supply of Indian workers exerts downward pressure on both non-remote and remote US wages in all sectors. Because Indian workers are closer substitutes with remote than with non-remote US workers, the decline in remote wages is significantly more pronounced. Remote wages fall most in the Web sector (-1.9%) and least in the Writing sector (-0.7%), which are respectively the sectors with the largest and the smallest shares of Indian workers in the remote wage bill. Non-remote wages fall most in the Web (-0.7%) and Design (-0.6%) sectors, which are the sectors with the largest share of Indian labor in the total wage bill. Given our estimate of the supply elasticity θ, these wage pressures trigger substantial reallocation of US workers from remote to non-remote work, resulting in a 3.6% to 8.1% decline in remote labor sourced from the US.

Finally, the increased supply of Indian workers lowers the dollar price of sectoral goods P_j. The implied dollar value of this price decline for US consumers corresponds to an increase of just 0.06% in their initial expenditures, reflecting both India's modest cost share in the total (remote + non-remote) wage bills of the four sectors and the fact that these four sectors account for only 20% of total expenditures in our calibration (γ_j in Table 4). Consequently, US remote workers are worse off as a result of the increased supply of Indian remote workers. Non-remote workers in the Design and Web sectors also face net losses, while in Writing and Admin sectors the price decline more than compensates for the fall in wages.

5.4 Increase in the supply of remote labor from the US

The final counterfactual considers an increase in the supply of remote labor from the US, motivated by the Broadband Equity, Access, and Deployment (BEAD) federal program, which aims to expand high-speed internet access in rural and underserved areas. Recent estimates suggest this initiative could expand high-speed access to roughly 7.6 million additional people in the US, which corresponds to an increase of 2.5% in the fraction of workers with access to broadband internet.[29] We assume newly connected workers

[28] See the official press release here for additional details.

[29] Currently, 93% of the population has access to broadband. The US population is 342 million, so the fraction of the US population that will gain access to broadband internet under the policy is $7.6/342 = 2.3\%$.

would, at the same wages, participate in remote work at the same rate as currently connected workers. We target a change in the cost of remote work, κ_{jc}, so that, at pre-shock wages, the fraction of workers that choose to work remotely increases by 2.5% (i.e., we set a higher $\hat{\kappa}^{\theta}_{jc} = 1.025$ for $c = \text{US}$).

The last panel in Table 5 reports the result of this counterfactual. The increase in the supply of US remote workers leads to a reduction in US remote wages, while the policy-induced decline in the supply of non-remote labor puts upward pressure on US non-remote wages. The decline in remote wages is roughly 0.1% across all four sectors (see s_{jc} for $c = \text{US}$ in Table 4). We note that, although US remote wages decline, remote workers' net income ($\hat{\kappa}_{jU}\hat{\Omega}_{jU}$) increases by 0.4% to 0.6%. Non-remote wages increase modestly in all sectors, by less than 0.1%. Interestingly, approximately half of the expansion in the US remote wage bill arises from the displacement of foreign remote workers.[30] This suggests that an expansion of the US remote labor supply would reshore work currently performed abroad. Finally, the prices of the sectoral goods increase by 0.007% in Admin, 0.034% in Design, 0.031% in Web, and 0.007% in Writing, which consumers value as a 0.003% decrease in total expenditures.

5.5 Robustness

This section evaluates the sensitivity of our results to the calibrated value of the elasticity of substitution ϵ, and assesses how our empirical estimates influence the model's predictions by varying the values of the supply elasticity θ, the substitution elasticity ρ, and the initial offshoring shares $1 - s_{jU}$. Rather than repeating each counterfactual, we analyze these sensitivities by focusing on our first counterfactual exercise, and focus our discussion on the Design sector (the most affected sector in this counterfactual). The patterns are similar in all four sectors.

Sensitivity to model parameters

Panels (ii) and (iii) in Appendix Table A12 report the results from our first counterfactual as we halve and double the baseline value of $\epsilon = 3.6$, recalibrating θ and ρ in each case to

This implies an increase in the number of people with internet of $\frac{93\%+2.3\%}{93\%} - 1 = 2.5\%$.

[30]Since total sectoral expenditures $\bar{\gamma}_j$ are fixed, the weighted changes in the non-remote, US remote, and foreign remote wage bills must sum to zero. Thus, the foreign displacement share is the ratio of the decline in the foreign remote wage bill to the expansion of the US remote wage bill, $-\frac{\left(\sum_{c \neq U} s_{jc}\hat{\Omega}_{jc}\hat{N}_{jc} - (1-s_{jU})\right)}{s_{jU}\left(\hat{\Omega}_{jU}\hat{N}_{jU} - 1\right)}$. The foreign displacement explains 55% in Admin, 44% in Design and Web, and 51% in Writing.

Table 5: Counterfactual Exercises

	Sector			
Counterfactual	Admin	Design	Web	Writing
i. Reshoring remote jobs				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	12.9	22.5	22.3	9.2
Non-Remote US wage: $B_{jU} - 1$ (%)	1.8	10.3	7.7	1.7
Remote US labor: $\hat{N}_{jU} - 1$ (%)	100.2	101.1	133.7	61.0
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-0.9	-5.1	-4.2	-0.8
Compensating variation: $\sum_j \gamma_j \ln \hat{P}_j$	0.83			
ii. Increase in the demand for remote labor				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	8.9	4.7	6.8	8.4
Non-Remote US wage: $B_{jU} - 1$ (%)	-2.0	-6.6	-4.5	-2.4
Remote US labor: $\hat{N}_{jU} - 1$ (%)	101.7	115.0	110.9	100.8
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-0.9	-5.8	-3.5	-1.4
Compensating variation: $\sum_j \gamma_j \ln \hat{P}_j$	-0.91			
iii. Increase in supply of Indian remote workers				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	-0.9	-1.2	-1.9	-0.7
Non-Remote US wage: $B_{jU} - 1$ (%)	-0.1	-0.6	-0.7	-0.1
Remote US labor: $\hat{N}_{jU} - 1$ (%)	-4.9	-4.3	-8.1	-3.6
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	0.0	0.2	0.3	0.0
Compensating variation: $\sum_j \gamma_j \ln \hat{P}_j$	-0.064			
iv. Increase in supply of US remote workers				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	-0.1	-0.1	-0.1	-0.1
Non-Remote US wage: $B_{jU} - 1$ (%)	0.0	0.0	0.0	0.0
Remote US labor: $\hat{N}_{jU} - 1$ (%)	1.6	1.5	1.6	1.5
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	0.0	-0.1	-0.1	0.0
Compensating variation: $\sum_j \gamma_j \ln \hat{P}_j$	0.003			

match the same empirical moments. Halving ϵ raises the predicted increase in US remote wages in the Design sector from 22.5% to 30.8%, and amplifies the expansion of US remote employment from 101.1% to 126.4%. Doubling ϵ reduces these effects to 17.6% and 72.7%, respectively. In contrast, non-remote wages are largely insensitive to ϵ, ranging between 9.5% and 10.7% across these calibrations, suggesting that the indirect equilibrium channel through which remote shocks affect non-remote wages is quite stable. The intuition is that a lower ϵ implies remote and non-remote labor are less substitutable. When foreign remote workers are removed from the market, it is harder to shift to non-remote labor, which results in larger wage and employment effects in the remote labor market.

Panels (iv) and (v) each vary a single parameter from the baseline: panel (iv) halves the elasticity of remote labor supply, θ, and panel (v) doubles the elasticity of substitution across foreign nationalities, ρ. Both changes imply a coefficient $\frac{\theta}{\rho+\theta} = 0.12$, which falls in the middle of the range we estimate for the ERPT elasticity using ask wage data. Halving θ increases the remote wage gain in the Design sector from 22.5% to 28.5%, while reducing the expansion of US remote labor from 101.1% to 69.2%. With a less elastic labor supply, the equilibrium adjusts to reshoring primarily through higher wages rather than higher quantities. Doubling ρ relative to its baseline of 24.6 yields US remote wage gains of 23.2%, little changed from the baseline. This insensitivity is intuitive: at $\rho = 24.6$, remote workers from different countries are already close substitutes, so further increases in ρ have a limited effect on the counterfactual results.

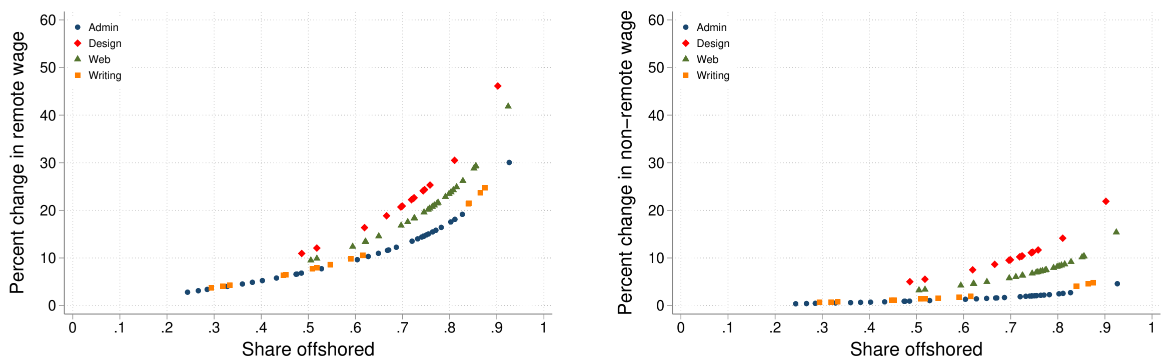
Overall, the baseline findings are preserved: reshoring foreign remote work raises both US remote and non-remote wages, expands US remote employment, and modestly contracts non-remote employment.

How does offshoring of remote work impact US wages?

In our first counterfactual, we demonstrated that the wage effects of reshoring all remote work are more pronounced in sectors where a large fraction of the wage bill is offshored. Although our quantitative model was initially implemented across the four aggregate sectors used in the estimation in Section 3.3, we now evaluate wage changes across the 91 detailed occupations for which we can compute the offshorability measure presented in Section 3.1.

With this in mind, we study a 91-sector model with the shares α_j and $\frac{N_{jU}}{L_{jU}}$ assumed to be common across subsectors within the corresponding broad category. Importantly, the shares of the offshored wage bill are allowed to vary across these 91 subsectors to match the data in Section 3.1. We then simulate the reshoring of all remote labor to the US within this expanded model, as we did in our first counterfactual. Figure 5 illustrates the results. The increase in US remote wages ranges from 2.8% to 45% across subsectors (Panel A), with differences explained primarily by the share of the sectoral wage bill currently offshored. For example, wage gains from reshoring remote work would be largest for Technical Support Representatives, ERP/CRM Specialists, and Motion Graphics Workers, and would be smallest for Grant Writers, Contract Law Professionals, and Corporate Law Professionals. The effect on non-remote wages depicted in Panel B is smaller across the board but follows a similar pattern.

Figure 5: Effects of Labor Offshoring on US Wages



Notes: Each dot represents one of the 91 detailed occupations on the platform. The x-axis reports the share of the occupation's US wage bill performed by non-US workers. The y-axis in each panel reports the predicted percent change in US remote wages (Panel A) and US non-remote wages (Panel B) from a counterfactual that eliminates all foreign remote labor. Colors indicate the broad sector to which each occupation belongs.

6 Conclusion

This paper uses novel data from a large web-based job platform to study how the price of remote work is determined in a globalized labor market. Despite the global nature of the platform, we find large wage gaps that are strongly correlated with the GDP per capita of the workers' country, and are not accounted for by differences in workers' characteristics, occupations, or employers' locations. Data on wage changes suggest that this correlation is driven by differences in the wages and prices that remote workers face in their local labor markets. We also document that remote wages in local currency move with the dollar exchange rate of the worker's country, and are highly sensitive to changes in the wages of foreign competitors. Together, these findings indicate that remote labor markets are globally competitive, yet remain anchored to local outside options.

We develop and calibrate a model that rationalizes these empirical patterns and use it to quantify the implications of global remote competition. We find that eliminating foreign remote labor would raise US wages in exposed sectors by up to 22.2%, while expansions in foreign remote labor supply exert downward pressure on US remote wages and have smaller but still negative effects on non-remote wages. Policies that instead expand US participation in remote work would reshore approximately half of the expansion from the displacement of foreign remote workers.

Overall, our results suggest that the rise of remote work is reshaping the way labor market shocks are transmitted across borders, integrating service markets in ways that parallel the globalization of goods trade while preserving an important role for local labor market conditions. As remote work continues to expand, several questions remain open. Future research could study how firms adjust their organization and task allocation in response to global remote competition, and how the distributional consequences of remote work differ across skill groups. Understanding these dynamics will be central for assessing the long-run implications of remote work for wage inequality, offshoring risk, and the design of labor and trade policy.

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Online Appendix

A.1 Additional Tables and Figures

Table A1: List of Occupations

Detailed Occupation	Broad Occ.	Detailed Occupation	Broad Occ.
Accounting Freelancers	Accounting	Brand Identity Strategy Freelancers	Design
Financial Planners & Advisors	Accounting	Graphics Design Freelancers	Design
HR & Recruiting Professionals	Accounting	Logo & Brand Designers	Design
Management Consultants	Accounting	Motion Graphics Freelancers	Design
Other - Accounting & Consulting Specialists	Accounting	Other - Design & Creative	Design
Data Entry Specialists	Admin	Photographers	Design
Other - Admin Support Professionals	Admin	Physical Design Freelancers	Design
Project Managers	Admin	Presentation Designers & Developers	Design
Transcription Services Professionals	Admin	Video Production Specialists	Design
Virtual Assistants, Personal Assistants	Admin	Voice Talent Artists	Design
Web Research Specialists	Admin	3D Modeling CAD Freelancers	Engineering
Customer Service & Tech Support Reps	Customer Service	Architects	Engineering
Other - Customer Service Specialists	Customer Service	Chemical Engineers	Engineering
Technical Support Representatives	Customer Service	Contract Manufacturers	Engineering
A/B Testing Specialists	Data Science	Electrical Engineers	Engineering
Data Extraction / ETL Specialists	Data Science	Interior Designers	Engineering
Data Mining Management Freelancers	Data Science	Mechanical Engineers	Engineering
Data Visualization Specialists & Analysts	Data Science	Other - Engineering & Architecture Specialists	Engineering
Machine Learning Specialists & Analysts	Data Science	Product Designers	Engineering
Other - Data Science & Analytics Professionals	Data Science	Structural & Civil Engineers	Engineering
Quantitative Analysis Specialists	Data Science	Database Administration Freelancers	IT
Animators	Design	ERP / CRM Implementation Specialists	IT
Art Illustration Freelancers	Design	Information Security Specialists & Consultants	IT
Audio Production Specialists	Design	Network & System Administrators	IT
		Other - IT & Networking	IT

Table A1: List of Occupations (continued)

Detailed Occupation	Broad Occ.	Detailed Occupation	Broad Occ.
Contract Law Freelancers	Legal	Desktop Software Developers	Web & soft.
Corporate Law Professionals & Consultants	Legal	E-commerce Programmers & Developers	Web & soft.
Criminal Law Professionals & Consultants	Legal	Game Developers	Web & soft.
Family Law Professionals & Consultants	Legal	Mobile Developers	Web & soft.
Intellectual Property Law Professionals & Consultants	Legal	Other Software Development Freelancers	Web & soft.
Other Legal Freelancers	Legal	Product Management Professionals & Consultants	Web & soft.
Paralegal Professionals	Legal	QA & Testing Specialists	Web & soft.
Display Advertising Specialists	Sales	Scripts & Utilities Developers	Web & soft.
Email & Marketing Automation Managers & Consultants	Sales	Web Designers, Mobile Designers	Web & soft.
Lead Generation Professionals	Sales	Web Developers	Web & soft.
Market Researchers, Customer Researchers	Sales	Academic Writers & Researchers	Writing
Marketing Strategy Freelancers	Sales	Article Blog Writing Freelancers	Writing
Other Sales & Marketing Specialists	Sales	Copywriters	Writing
Public Relations (PR) Professionals	Sales	Creative Writers	Writing
Search Engine Marketing (SEM) Specialists	Sales	Grant Writers	Writing
Search Engine Optimization (SEO) Specialists	Sales	Other Writing Services Professionals	Writing
Social Media Marketing (SMM) Specialists	Sales	Proofreaders & Editors	Writing
Telemarketing & Telesales Specialists	Sales	Resumes & Cover Letters Writers	Writing
General Translation Freelancers	Translation	Technical Writers	Writing
Legal Translation Professionals	Translation	Web Content Writers, Web Content Managers	Writing
Medical Translation Professionals	Translation		
Technical Translation Professionals	Translation		

Table A2: Concordance between Platform Occupations and SOC Classification

Platform Occupation	SOC code	SOC title	Platform Occupation	SOC code	SOC title
3D Modeling CAD Freelancers	27-1014	Special Effects Artists and Animators	Logo & Brand Designers	27-1024	Graphic Designers
A/B Testing Specialists	15-1250	Software and Web Developers, Programmers	Machine Learning Specialists & Analysts	15-2051	Data Scientists
Accounting Freelancers	13-2011	Accountants and Auditors	Management Consultants	13-1111	Management Analysts
Animators	27-1014	Special Effects Artists and Animators	Market Researchers, Customer Researchers	13-1161	Market Research Analysts and Mktg Spec.
Architects	17-1011	Architects, Except Landscape and Naval	Marketing Strategy Freelancers	13-1161	Market Research Analysts and Mktg Spec.
Art Illustration Freelancers	27-1013	Fine Artists	Mechanical Engineers	17-2141	Mechanical Engineers
Article Blog Writing Freelancers	27-3043	Poets, Lyricists and Creative Writers	Medical Translation Professionals	27-3091	Interpreters and Translators
Audio Production Specialists	27-4011	Audio and Video Technicians	Mobile Developers	15-1252	Software Developers
Chemical Engineers	17-2041	Chemical Engineers	Network & System Administrators	15-1244	Network and Computer Systems Admin.
Contract Law Freelancers	23-1011	Lawyers	Other - Admin Support Professionals	43-4151	Order Clerks
Contract Manufacturers	17-3011	Architectural and Civil Drafters	Other Sales & Marketing Specialists	13-1161	Search Marketing Strategists
Copywriters	27-3043	Writers and Authors	Other Writing Services Professionals	27-3043	Writers and Authors
Corporate Law Professionals & Consultants	23-1011	Lawyers	Paralegal Professionals	23-2011	Paralegals and Legal Assistants
Creative Writers	27-3043	Poets, Lyricists and Creative Writers	Photographers	27-4021	Photographers
Customer Service & Tech Support Reps	43-4051	Customer Service Representatives	Presentation Designers & Developers	27-1011	Art Directors
Data Entry Specialists	43-9021	Data Entry Keyers	Product Management Professionals & Consultants	13-1081	Logistics Analysts
Data Extraction / ETL Specialists	15-1243	Data Warehousing Specialists	Project Managers	13-1082	Project Management Specialists
Data Mining Management Freelancers	15-2051	Data Scientists	Proofreaders & Editors	27-3041	Editors
Data Visualization Specialists & Analysts	15-2051	Data Scientists	Public Relations (PR) Professionals	27-3031	Public Relations Specialists
Database Administration Freelancers	15-1242	Database Administrators	QA & Testing Specialists	15-1253	Software Quality Assurance Analysts
Desktop Software Developers	15-1252	Software Developers	Quantitative Analysis Specialists	15-2051	Data Scientists
Display Advertising Specialists	13-1161	Search Marketing Strategists	Resumes & Cover Letters Writers	21-1012	Educational, Guidance, and Career Counselors
ERP / CRM Implementation Specialists	15-1211	Computer Systems Analysts	Scripts & Utilities Developers	15-1251	Computer Programmers
E-commerce Programmers & Developers	13-1161	Search Marketing Strategists	Search Engine Marketing (SEM) Specialists	13-1161	Search Marketing Strategists
Electrical Engineers	17-2071	Electrical Engineers	Search Engine Optimization (SEO) Specialists	13-1161	Market Research Analysts and Mktg Spec.
Email & Marketing Automation Managers & Consultants	13-1161	Search Marketing Strategists	Social Media Marketing (SMM) Specialists	13-1161	Search Marketing Strategists
Family Law Professionals & Consultants	23-1011	Lawyers	Technical Support Representatives	15-1232	Computer User Support Specialists
Game Developers	15-1255	Video Game Designers	Technical Translation Professionals	27-3091	Interpreters and Translators
General Translation Freelancers	25-1124	Foreign Lang. and Literature Teachers, PSE	Technical Writers	27-3042	Technical Writers
Grant Writers	13-1131	Fundraisers	Telemarketing & Telesales Specialists	41-9041	Telemarketers
Graphics Design Freelancers	27-1024	Graphic Designers	Transcription Services Professionals	27-4011	Audio and Video Technicians
Information Security Specialists & Consultants	15-1212	Information Security Analysts	Video Production Specialists	27-2012	Producers and Directors
Intellectual Property Law Professionals & Consultants	23-1011	Lawyers	Virtual Assistants, Personal Assistants	27-1014	Special Effects Artists and Animators
Interior Designers	27-1025	Interior Designers	Voice Talent Artists	27-2042	Musicians and Singers
Lead Generation Professionals	11-2021	Marketing Managers	Web Designers, Mobile Designers	15-1255	Web and Digital Interface Designers
Legal Translation Professionals	27-3091	Interpreters and Translators	Web Research Specialists	15-2051	Data Scientists

Table A3: Offshoring by Occupation

Platform Occupation	Value share offshored	Quantity share offshored	Platform Occupation	Value share offshored	Quantity share offshored
Technical Support Representatives	0.93	0.89	Presentation Designers & Developers	0.72	0.82
ERP / CRM Implementation Specialists	0.92	0.95	Architects	0.71	0.89
Motion Graphics Freelancers	0.90	0.89	Video Production Specialists	0.70	0.81
Medical Translators Professionals	0.88	0.90	Logo & Brand Designers	0.70	0.81
Legal Translation Professionals	0.86	0.88	Data Mining Management Freelancers	0.70	0.89
Other Software Development Freelancers	0.86	0.90	Photographers	0.70	0.83
Mobile Developers	0.86	0.90	Project Managers	0.69	0.79
Interior Designers	0.85	0.90	Email & Marketing Automation Managers & Consultants	0.67	0.80
Technical Translation Professionals	0.84	0.87	Market Researchers, Customer Researchers	0.67	0.83
General Translation Freelancers	0.84	0.88	Audio Production Specialists	0.67	0.76
Machine Learning Specialists & Analysts	0.83	0.89	Mechanical Engineers	0.65	0.81
Virtual Assistants, Personal Assistants	0.83	0.88	HR & Recruiting Professionals	0.65	0.70
QA & Testing Specialists	0.81	0.80	Other - Data Science & Analytics Professionals	0.63	0.79
Web Research Specialists	0.81	0.87	Data Visualization Specialists & Analysts	0.62	0.77
Animators	0.81	0.89	Contract Manufacturers	0.62	0.80
Network & System Administrators	0.81	0.87	Chemical Engineers	0.62	0.71
Information Security Specialists & Consultants	0.80	0.88	Other - Engineering & Architecture Specialists	0.62	0.77
Data Entry Specialists	0.80	0.88	Technical Writers	0.62	0.64
Desktop Software Developers	0.80	0.87	Marketing Strategy Freelancers	0.60	0.76
Ecommerce Programmers & Developers	0.79	0.85	Electrical Engineers	0.59	0.80
Other - Customer Service Specialists	0.78	0.84	Copywriters	0.59	0.61
Scripts & Utilities Developers	0.78	0.81	Proofreaders & Editors	0.55	0.56
Product Management Professionals & Consultants	0.77	0.84	Accounting Freelancers	0.53	0.68
Family Law Professionals & Consultants	0.77	0.56	Article Blog Writing Freelancers	0.52	0.57
Web Developers	0.77	0.72	A/B Testing Specialists	0.52	0.76
Customer Service & Tech Support Reps	0.76	0.83	Voice Talent Artists	0.52	0.55
3D Modeling Cad Freelancers	0.76	0.87	Web Content Writers, Web Content Managers	0.51	0.59
Structural & Civil Engineers	0.76	0.88	Quantitative Analysis Specialists	0.51	0.70
Other - IT & Networking	0.76	0.86	Physical Design Freelancers	0.49	0.86
Other - Admin Support Professionals	0.75	0.87	Display Advertising Specialists	0.49	0.64
Web Designers, Mobile Designers	0.75	0.84	Other - Accounting & Consulting Specialists	0.48	0.69
Search Engine Marketing (SEM) Specialists	0.75	0.83	Criminal Law Professionals & Consultants	0.47	0.66
Brand Identity Strategy Freelancers	0.75	0.78	Creative Writers	0.45	0.48
Data Extraction / ETL Specialists	0.75	0.87	Other Writing Services Professionals	0.45	0.51
Transcription Services Professionals	0.75	0.78	Financial Planners & Advisors	0.43	0.62
Telemarketing & Telesales Specialists	0.74	0.85	Paralegal Professionals	0.40	0.36
Social Media Marketing (SMM) Specialists	0.74	0.86	Public Relations (PR) Professionals	0.38	0.57
Product Designers	0.74	0.89	Management Consultants	0.36	0.53
Graphics Design Freelancers	0.74	0.82	Resumes & Cover Letters Writers	0.33	0.35
Search Engine Optimization (SEO) Specialists	0.74	0.83	Intellectual Property Law Professionals & Consultants	0.33	0.38
Other Sales & Marketing Specialists	0.73	0.84	Academic Writers & Researchers	0.32	0.37
Game Developers	0.73	0.88	Grant Writers	0.29	0.30
Other - Design & Creative	0.72	0.83	Contract Law Freelancers	0.29	0.33
Database Administration Freelancers	0.72	0.84	Other Legal Freelancers	0.27	0.38
Art Illustration Freelancers	0.72	0.79	Corporate Law Professionals & Consultants	0.24	0.33
Lead Generation Professionals	0.72	0.87			

Table A4: Offshoring by SOC Occupation

SOC code	SOC title	Value share offshored	Quantity share offshored	SOC code	SOC title	Value share offshored	Quantity share offshored
15-1232	Computer User Support Specialists	0.93	0.89	27-1011	Art Directors	0.72	0.82
15-1211	Computer Systems Analysts	0.92	0.95	13-1161	Search Marketing Strategists	0.72	0.82
27-3091	Interpreters and Translators	0.85	0.88	13-1161	Market Research Analysts and Marketing Specialists	0.72	0.82
27-1025	Interior Designers	0.85	0.90	17-1011	Architects, Except Landscape and Naval	0.71	0.89
25-1124	Foreign Language and Literature Teachers, Postsecondary	0.84	0.88	27-2012	Producers and Directors	0.70	0.81
15-1252	Software Developers	0.83	0.89	27-4021	Photographers	0.70	0.83
15-1253	Software Quality Assurance Analysts and Testers	0.81	0.80	13-1082	Project Management Specialists	0.69	0.79
27-1014	Special Effects Artists and Animators	0.81	0.88	17-2141	Mechanical Engineers	0.65	0.81
15-1244	Network and Computer Systems Administrators	0.81	0.87	17-3011	Architectural and Civil Drafters	0.62	0.80
15-1212	Information Security Analysts	0.80	0.88	17-2041	Chemical Engineers	0.62	0.71
43-9021	Data Entry Keyers	0.80	0.88	27-3042	Technical Writers	0.62	0.64
15-1251	Computer Programmers	0.78	0.81	17-2071	Electrical Engineers	0.59	0.80
13-1081	Logistics Analysts	0.77	0.84	27-3041	Editors	0.55	0.56
43-4051	Customer Service Representatives	0.76	0.83	13-2011	Accountants and Auditors	0.53	0.68
43-4151	Order Clerks	0.75	0.87	15-1250	Software and Web Developers, Programmers, and Testers	0.52	0.76
15-1255	Video Game Designers	0.75	0.84	27-2042	Musicians and Singers	0.52	0.55
15-1255	Web and Digital Interface Designers	0.75	0.84	27-3043	Writers and Authors	0.50	0.55
15-1243	Data Warehousing Specialists	0.75	0.87	27-3043	Poets, Lyricists and Creative Writers	0.50	0.55
41-9041	Telemarketers	0.74	0.85	23-2011	Paralegals and Legal Assistants	0.40	0.36
15-2051	Data Scientists	0.74	0.85	23-1011	Lawyers	0.39	0.40
27-1024	Graphic Designers	0.73	0.81	27-3031	Public Relations Specialists	0.38	0.57
15-1242	Database Administrators	0.72	0.84	13-1111	Management Analysts	0.36	0.53
27-1013	Fine Artists, Including Painters, Sculptors, and Illustrators	0.72	0.79	21-1012	Educational, Guidance, and Career Counselors and Advisors	0.33	0.35
27-4011	Audio and Video Technicians	0.72	0.77	13-1131	Fundraisers	0.29	0.30
11-2021	Marketing Managers	0.72	0.87				

Table A5: Wage Determinants

	Coef.	Std. Err.			Coef.	Std. Err.
Experience				**Quality ratings**		
Earnings (in logs)	0.0723***	(0.00175)		Top rated	0.132***	(0.0048)
<=5 jobs	-0.0424***	(0.00578)		SR <70%	-0.167***	(0.0229)
[6,15) jobs	-0.0610***	(0.00625)		SR [70%,80%)	-0.0745***	(0.0165)
[15,50) jobs	-0.0390***	(0.00771)		SR [80%,90%)	-0.0773***	(0.0130)
>=50 jobs	-0.00258	(0.0172)		SR [90%,95%)	-0.0497***	(0.0128)
Part time/full time				SR [95%,100%)	-0.0380***	(0.0124)
As needed	0.141***	(0.0108)		SR 100%	-0.100***	(0.0120)
<= 30 hrs/week	0.0982***	(0.0117)		**Skills**		
> 30 hrs/week	0.0779***	(0.0105)		Number of tests	-0.0018***	(0.0003)
Response time				Av. score	0.0581***	(0.00542)
< 24 hrs	-0.0415***	(0.00861)		**Agency**		
24 hrs - 3 days	0.0781***	(0.00507)		Single worker	0.148***	(0.0125)
3+ days	0.0572***	(0.0145)		Multi worker	-0.0437***	(0.0134)
Observations	90,550	R^2	0.551			

Notes: The table reports the coefficients estimated from equation (2). The sample consists of worker-employer pairs with available transacted wage data. *: significant at the 10% level, **: significant at the 5% level, ***: significant at the 1% level.

Table A6: Variance Decomposition of Wages

Component	Share of variance
Country of worker	0.23
Country of employer	0.004
Controls	0.17
Cov (country of worker, controls)	0.15
Cov (country of employer, controls)	0.0008
Cov (country of employer, country of worker)	0.002
Residual	0.45

Notes: The table reports the variance decomposition of equation (2) using transacted wages. Row (1) reports the variance accounted for by the country of worker $\mathbb{C}_i$; row (2), by the country of employer $\mathbb{D}_f$; and row (3), by the controls $\mathbb{I}_{i=f}$ and $\boldsymbol{\beta}'\boldsymbol{X_i}$ jointly. Rows (4) and (5) show two times the covariance between $\mathbb{C}_i$ and controls and between $\mathbb{D}_f$ and controls, respectively. Row (6) shows two times the covariance between $\mathbb{C}_i$ and $\mathbb{D}_f$. Row (7) is the variance not explained.

Table A7: Frequency of Transacted Wage Changes

Sample	Freq. Wage Changes	Share Wage Increases	Med. Wage Increase	Med. Wage Decrease
All	0.76	0.64	0.26	-0.22
$\Delta T = 1$	0.69	0.58	0.22	-0.22
$\Delta T \leq med(\Delta T)$	0.71	0.60	0.22	-0.22
$\Delta T > med(\Delta T)$	0.82	0.69	0.29	-0.22

Notes: The table presents summary statistics on the distribution of within-worker changes in transacted wages between consecutive hourly jobs.

Table A8: Pass-through to Transacted Wages: First Stage

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$\Delta_s w_{jt}$	$\Delta_s w_{jt}$	$\Delta_s w_{jt}$	$\Delta_s w_{jt}$	$\Delta_s w_{jt}$	$\Delta_s w_{jt}^{plac}$	$\Delta_s w_{jt}$	$\Delta_s w_{jt}$
$\Delta_s e_{ct}$	-0.005		-0.007	-0.005	-0.005	0.000	-0.023	0.036*
	(0.018)		(0.018)	(0.018)	(0.018)	(0.003)	(0.025)	(0.020)
π_{c,t_s}	-0.020		-0.019	-0.019	-0.020	-0.003	-0.068*	0.081
	(0.028)		(0.027)	(0.027)	(0.028)	(0.015)	(0.038)	(0.067)
$\pi_{c,t_s} + \Delta_s e_{ct}$		-0.005						
		(0.018)						
$\Delta_s e_{jt}$	0.728***	0.731***	0.782***	0.727***	0.728***	0.061**	0.916***	0.042**
	(0.096)	(0.099)	(0.091)	(0.098)	(0.096)	(0.030)	(0.091)	(0.020)
$\Delta_s e_{jt}^{plac}$					0.000	0.018***		
					(0.000)	(0.002)		
Observations	88399	88399	66526	88399	88399	88399	88399	88399

Notes: Column 1 reports the first stage corresponding to Column 3 in Table 3. Columns 2-4 report the first stage corresponding to Columns 2-4 in Table A9. Columns 5-6 report the first stage corresponding to Column 5 in Table A9. Columns 7-8 report the first stage corresponding to Columns 6-7 in Table A9. Specifications in these columns include country-sector-specific linear trends but they are not reported. Standard errors are two-way clustered at the sector-time-spell and country levels. *: significant at the 10% level, **: significant at the 5% level, ***: significant at the 1% level.

Table A9: Pass-through to Transacted Wages: Robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$	$\Delta_s w_{ijt}$
$\Delta_s e_{ct}$			0.251***	0.214***	0.214***	0.186**	0.376	0.084***	0.136***
			(0.066)	(0.053)	(0.053)	(0.088)	(0.284)	(0.028)	(0.044)
π_{c,t_s}			0.239*	0.195*	0.195*	0.204	0.516	0.095	0.005
			(0.137)	(0.104)	(0.104)	(0.218)	(0.578)	(0.086)	(0.141)
$\pi_{c,t_s} + \Delta_s e_{ct}$	0.203***	0.214***							
	(0.058)	(0.052)							
$\Delta_s w_{jt}$		0.727***	0.797***		0.716***	0.896***	-3.470		
		(0.263)	(0.282)		(0.256)	(0.154)	(4.341)		
$\Delta_s w_{-ijt}$				0.722***					
				(0.255)					
$\Delta_s w_{jt}^{plac}$					0.079				
					(0.122)				
Observations	88399	88399	66526	88399	88399	88399	88399	226559	117089
Test $\beta_1 = \beta_2$			0.93	0.84	0.84	0.93	0.66	0.90	0.34
Specification	OLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS	OLS	OLS
F stat 1st stage		54.0	73.5	55.1	58.4	101.9	4.20		

Notes: Columns 1-2 reestimate Columns 1 and 3 from Table 3 imposing the restriction that $\beta_1 = \beta_2$. Column 3 reestimates Column 3 in Table 3 using the sample of non-zero wage changes. Column 4 reestimates Column 3 in Table 3 replacing the baseline wage index $\Delta_s W_{jt}$ with the leave-one-out wage index $\Delta_s W_{-ijt} \equiv \sum_{l \neq i} \frac{s_{ljt}}{1-s_{ijt}} \Delta_s w_{lt} = \left[\Delta_s W_{jt} - s_{ijt}\Delta_s w_{ijt}\right] / \left[1 - s_{ijt}\right]$. This alternative specification alleviates the concern that the aggregate wage index $\Delta_s W_{jt}$ is by definition a function of each worker's wage, and is thus correlated with the error term. Column 5 reestimates Column 3 in Table 3 and includes the change in wages of workers that are predicted to be the least likely competitors of a given worker. These columns include country-sector-specific linear trends. Column 6 reestimates the specification in Column 3 of Table 3 without controlling for country-sector-specific trends. Column 7 reestimates the specification in Column 3 of Table 3 without controlling for time-spell fixed effects. In Columns 1-7, standard errors are two-way clustered at the sector-time-spell and country levels. Column 8 reports the results from estimating equation (6), and Column 9 uses the sample of non-zero wage changes. Standard errors are clustered at the country level. *: significant at the 10% level, **: significant at the 5% level, ***: significant at the 1% level. The corresponding first stage regressions are reported in Table A8.

Table A10: Concordance between Occupations in the CPS Data and Sectors in the Model

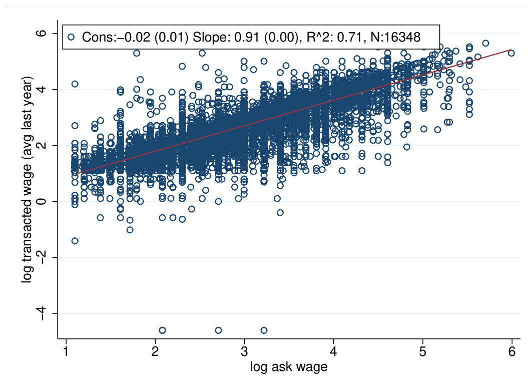
CPS code	CPS occupation description	Sector in the model	CPS code	CPS occupation description	Sector in the model
50	Marketing and sales managers	Admin	1540	Drafters	Web
430	Managers, all other	Admin	2000	Counselors	Writing
700	Logisticians	Web	2100	Lawyers	Admin
710	Management analysts	Admin	2145	Paralegals and legal assistants	Admin
726	Fundraisers	Writing	2200	Postsecondary teachers	Writing
735	Market research analysts and marketing specialists	Admin	2600	Artists and related workers	Design
740	Business operations specialists, all other	Admin	2630	Designers	Design
800	Accountants and auditors	Admin	2710	Producers and directors	Design
1006	Computer systems analysts	Web	2750	Musicians, singers, and related workers	Design
1007	Information security analysts	Web	2825	Public relations specialists	Admin
1010	Computer programmers	Web	2830	Editors	Writing
1030	Web developers	Web	2840	Technical writers	Writing
1060	Database administrators	Web	2850	Writers and authors	Writing
1105	Network and computer systems administrators	Web	2860	Miscellaneous media and communication workers	Writing
1050	Computer support specialists	Admin	2900	Broadcast and sound engineering technicians and radio operators	Design
1107	Computer occupations, all other	Web	2910	Photographers	Design
1240	Miscellaneous mathematical science occupations	Web	4940	Telemarketers	Admin
1300	Architects, except naval	Web	5240	Customer service representatives	Admin
1350	Chemical engineers	Web	5350	Order clerks	Admin
1410	Electrical and electronics engineers	Web	5810	Data entry keyers	Admin
1460	Mechanical engineers	Web			

Notes: There are 483 occupations in the CPS dataset. This table shows the CPS occupations that correspond to one of the four sectors: Admin, Design, Web, and Writing. The remaining CPS occupations, which are not included in this table, correspond to the outside sector in the model.

Table A11: List of Countries Included in the Model

United States	Finland	Latvia	Portugal
Albania	France	Lebanon	Qatar
Algeria	Georgia	Lithuania	Romania
Angola	Germany	Luxembourg	Saudi Arabia
Austria	Ghana	Malaysia	Senegal
Bangladesh	Greece	Maldives	Singapore
Barbados	Grenada	Mali	South Africa
Belgium	Guatemala	Malta	Spain
Benin	Guyana	Mauritius	Sri Lanka
Bolivia	Haiti	Mexico	Sweden
Bosnia and Herzegovina	Honduras	Montenegro	Switzerland
Brazil	Hungary	Morocco	Thailand
Bulgaria	Iceland	Namibia	Trinidad and Tobago
Cameroon	India	Nepal	Tunisia
Canada	Indonesia	Nicaragua	Turkey
Chile	Ireland	Nigeria	Uganda
Colombia	Israel	Norway	Ukraine
Costa Rica	Italy	Pakistan	United Arab Emirates
Cyprus	Jamaica	Panama	United Kingdom
Denmark	Japan	Paraguay	Uruguay
Ecuador	Jordan	Peru	Vietnam
El Salvador	Kenya	Philippines	

Figure A.1: Ask vs. transacted wages

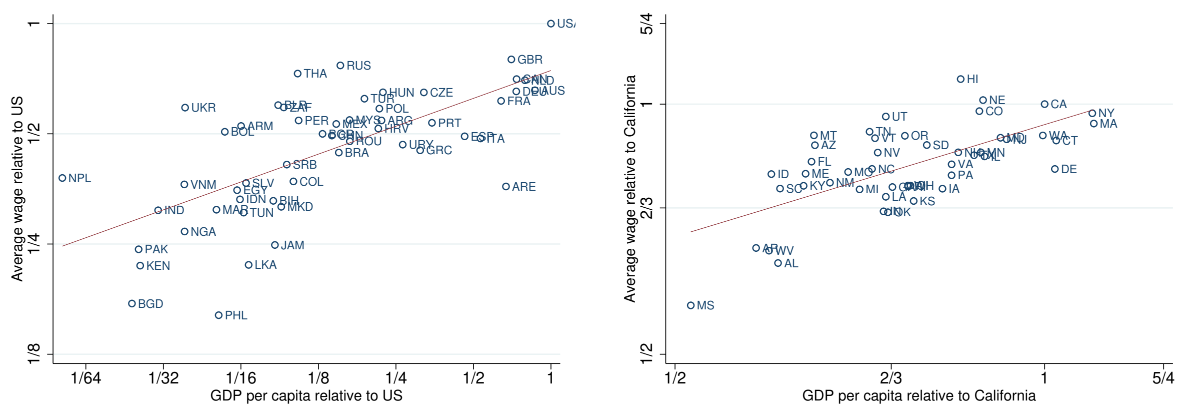


Notes: The figure shows a scatter plot of a worker's log average transacted wage (y-axis) against the worker's log ask wage (x-axis). Average transacted wages are computed using wages received within a one-year window centered on the date of the ask wage.

Figure A.2: Average wages (nonresidualized) across workers

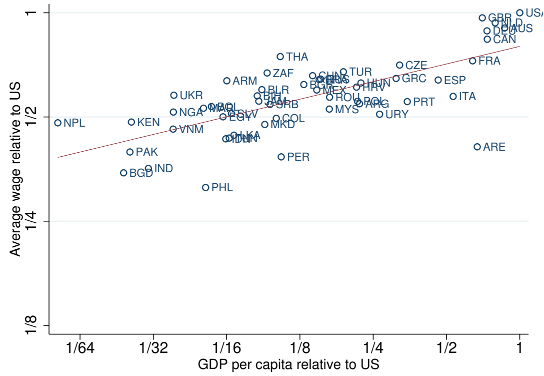
(a) By country

(b) By US state



Notes: The x-axis in panel (a) reports the (log of) the relative GDP per capita in US dollars, taken from the World Development Indicators (WDI). The y-axis plots the average (nonresidualized) transacted wage in each country relative to the US. The estimated slope is 0.25 (0.04) and the R^2 is 0.47. The x-axis in panel (b) reports the log of the relative GDP per capita in US dollars, taken from the Bureau of Economic Analysis. The y-axis plots the average ask wage in each state relative to California. The estimated slope is 0.44 (0.09) and the R^2 is 0.43. The red lines show the linear fit of the data.

Figure A.3: Average wages across workers: Ask wages

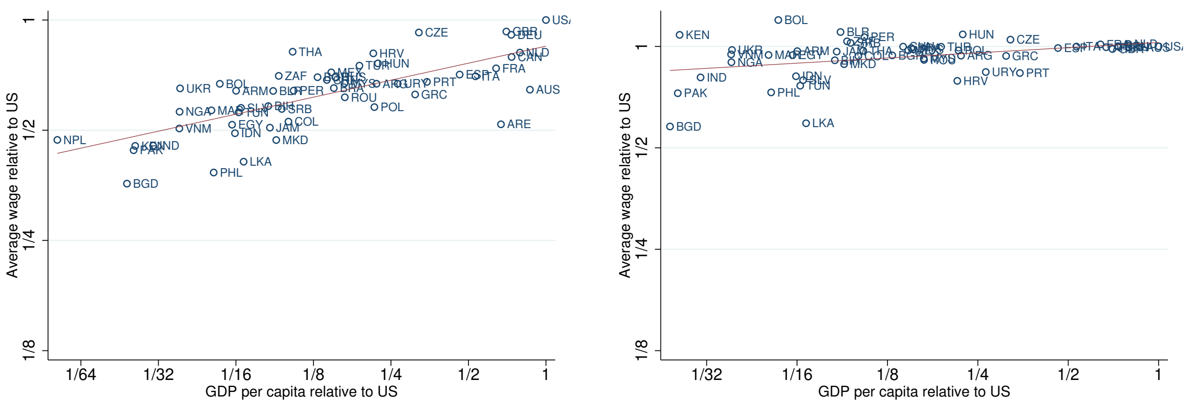


Notes: The x-axis reports the log of the relative GDP per capita in US dollars, taken from the World Development Indicators (WDI). The y-axis plots the residualized average wage in each country relative to the US obtained from the worker's country fixed effects estimated from a version of equation (2) that does not control for employers' country fixed effects. The outcome variable is ask wages, as opposed to transacted wages. The red lines show the linear fit of the data. The estimated slope is 0.17 (0.03) and the R^2 is 0.50.

Figure A.4: Differences in wages within workers and employers

(a) Within employer

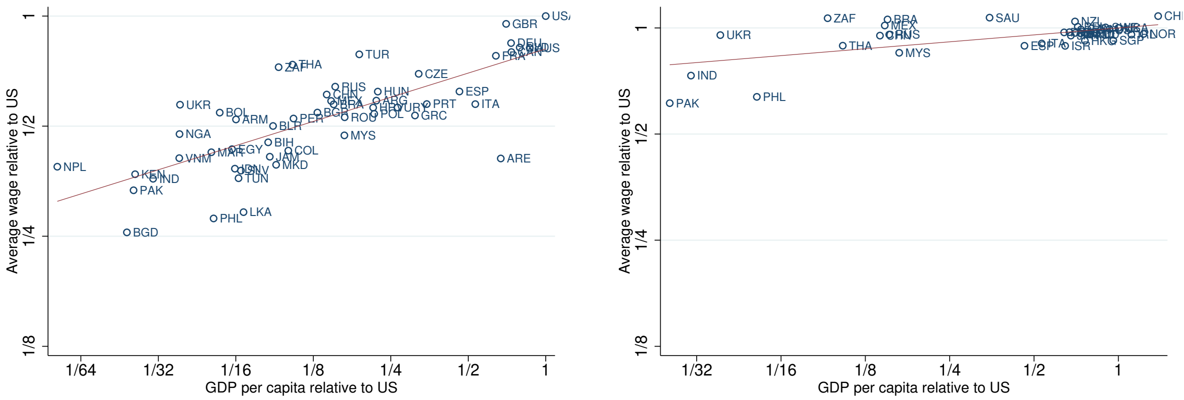
(b) Within worker



Notes: The x-axis reports the log of the relative GDP per capita in US dollars, taken from the World Development Indicators (WDI). The y-axis in panel (a) reports the set of country-of-worker fixed effects (relative to workers in the US) estimated according to a version of equation (2) that controls for employer fixed effects. The estimated slope is 0.15 (0.02) with an R^2 of 0.53. The y-axis in panel (b) reports the set of country-of-employer fixed effects (relative to employers in the US) estimated according to a version of equation (2) that controls for worker fixed effects. The estimated slope is 0.05 (0.02) with an R^2 of 0.14.

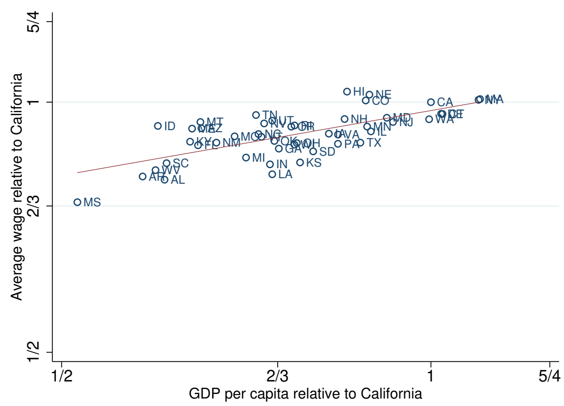
Figure A.5: Wages and GDP per capita relative to the US: controlling for distance

(a) Country of worker (b) Country of employer



Notes: The x-axes report the log of the relative GDP per capita in US dollars, taken from the World Development Indicators (WDI). The figure reports the average residualized wage in each country relative to the US obtained from the country-of-worker and country-of-employer fixed effects. These fixed effects are estimated according to equation (2) with the following additional control variables: dummy variables for whether the countries of the employer and worker are contiguous, share a common language, have colonial ties, share a common currency, and share a common legal origin. It also controls for the distance in kilometers between the capital cities of both countries weighted by the population size, and the time-zone difference (in hours) between the two countries. Panel (a) plots the worker's country fixed effects, and panel (b) plots the employer's country fixed effects. The estimated slope in panel (a) is 0.22 (0.03) and the R^2 is 0.58. The estimated slope in panel (b) is 0.07 (0.02) and the R^2 is 0.36.

Figure A.6: Wages and GDP per capita across US states (ask wages)



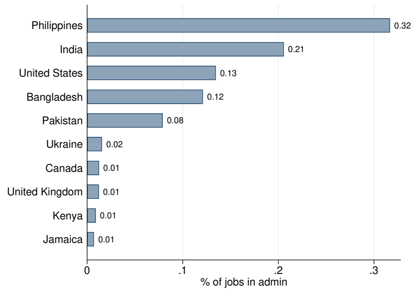
Notes: The x-axis reports the log of the relative GDP per capita in US dollars, taken from the Bureau of Economic Analysis. The figure plots the average ask wage in each state relative to California, obtained from state fixed effects in equation (A.4.1). The red line shows the linear fit of the data. The estimated slope is 0.26 (0.04) and the R^2 is 0.48.

Figure A.7: Real wages and comparison with non-remote wages

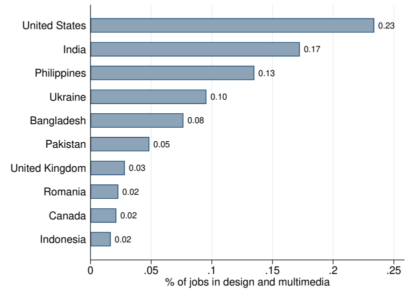
(a) Remote vs. non-remote wages (b) Remote wages vs. domestic price levels

Notes: In panel (a), the x-axis reports the average log compensation of employees in 2011 denominated in US dollars. The average compensation for each country is computed as the average among the following occupations included in the International Comparison Program (ICP) from the World Bank: Accounting and Bookkeeping Clerks, Human Resources Professionals, Computer Operators, Data Processing Managers, and Database Administrators. The y-axis reports the average residualized remote wage in each country relative to the US. In panel (b), the x-axis reports the price level of output included in the ICP (the ratio of the PPP exchange rate to the market exchange rate, PPP/XR; the US price level of output is normalized to 1 in 2017), relative to the US. The y-axis reports average residualized wages obtained from the country fixed effects estimated in equation (2). The red lines show the linear fit of the data. The estimated slope is 0.18 (0.04) and the R^2 is 0.41 in panel (a); the estimated slope is 0.52 (0.06) and the R^2 is 0.54 in panel (b).

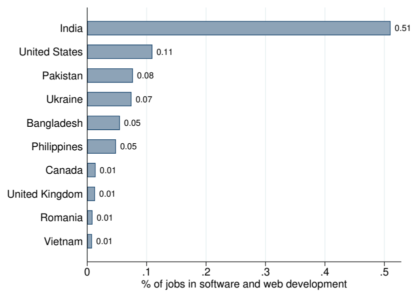
Figure A.8: Sectoral variation in instrumental variable



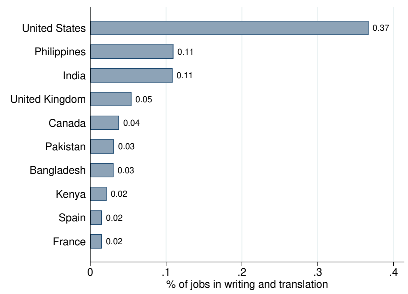
(a) Admin



(b) Design



(c) Web



(d) Writing

Notes: This figure reports the variation in the sectoral shares s_{jct} used to construct the instrumental variable $\sum_c s_{jct}\Delta_s e_{ct}$.

A.2 Data Appendix

Additional data sources: Our measure of GDP per capita in current US dollars is the variable gdp_pc_curr for year 2016 from the World Development Indicators (WDI). The GDP per capita in US dollars for each state is the variable SAGDP10N obtained from the US Department of Commerce, Bureau of Economic Analysis for year 2017. The "gravity" variables obtained from the CEPII Gravity Database are the following: contig, comlang_off, distw, tdiff, colony, comcur, comleg_pretrans, tradeflow_imf_d, gdp_ppp_o, and gdp_ppp_d (for a detailed description see http://www.cepii.fr/DATA_DOWNLOAD/gravity/doc/Gravity_documentation.pdf). For non-remote wages, we use the compensation of employees for year 2011 from the International Comparison Program (ICP) from the World Bank for the following occupations: Accounting and Bookkeeping Clerks, Human Resources Professionals, Computer Operators, Data Processing Managers, and Database Administrators. We adjust the value of compensation by the current exchange rate to convert it into US dollars. Finally, the exchange rate and inflation data used in Section 3 are sourced from the International Financial Statistics (IFS) database from the IMF.

Algorithm: The data on job history used in Section 3.3 record the sector for a subset of jobs. We assign sectors to the remaining jobs by applying a machine-learning algorithm to the information contained in the job descriptions. We first make the data suitable for analysis by removing a set of stop words (e.g., "and", "the"), punctuation marks, and numbers from the job description, which is available for all jobs. Then, we keep the 3,000 most frequent words, which balances the goal of using as many words as possible in the prediction step against the risk of overfitting the data. Next, we keep 70% of jobs with occupation data as a training sample, and use the remaining 30% as a validation sample. We then train an artificial neural network on the training sample using a hyperparameter optimization algorithm (see Chollet, 2021) to predict the broad occupation a given job belongs to based on the (cleaned) job description. To set the parameters of this algorithm, we perform a cross-validation exercise to achieve good prediction outcomes on the validation sample. Finally, we apply the estimated prediction model on the descriptions of jobs for which we do not have occupation data and obtain the likelihood that a given job belongs to each broad occupation. In our baseline analysis, we assign jobs to the occupation that obtains the highest likelihood.

Measurement of remote work and earnings: We use data from the May 2017 Current Population Survey (CPS) Contingent Worker Supplement, which includes questions designed to identify platform-based remote work. Specifically, the survey asks "*Some people select short, online tasks or projects through companies that maintain lists that are accessed through an app or a website. These tasks are done entirely online and the companies coordinate payment for the work (for example, data entry, translating text, web or software development, or graphic design). Does this describe any work you did last week?*" See https://www.census.gov/data/datasets/time-series/demo/cps/cps-supp_cps-repwgt/cps-contingent.html for details on the supplement. We use the variables PRERNWA (earnings), PRSOEM (indicator for work performed entirely online), and PWSSWGT (supplement weights). We replace zero earnings with the mean weekly earnings of workers in the same CPS occupation and mode of work (i.e., remote vs. non-remote). In a small number of occupations, earnings are missing for all remote workers. In those cases, we proceed as follows. We assume that the relative wage of non-remote workers to all workers within an occupation mirrors the corresponding relative wage in the aggregate CPS sample. Given that the average earnings in an occupation are a weighted average of remote and non-remote earnings, we use this assumption and the observed share of remote workers in each occupation to impute average remote earnings for that occupation accordingly.

A.3 Alternative measures of offshoring by occupation

A.3.1 Quantity-based measures

Section 3.1 measures the share of jobs that are offshored in terms of values. Here, we present an alternative measure that computes the share of jobs offshored by count (rather than by value). In particular, we compute

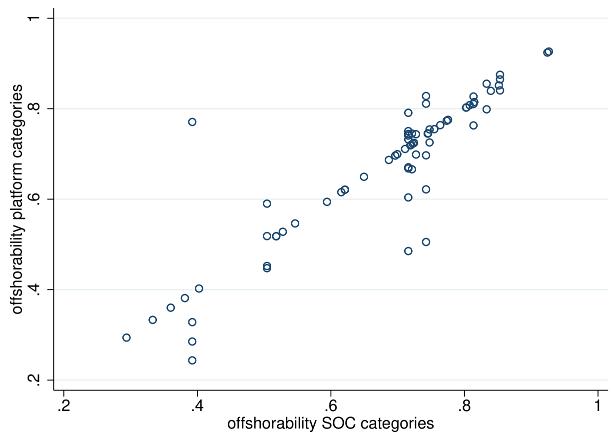
$$\widetilde{\mathcal{O}}_j = \frac{\text{jobs in } j \text{ where cty. employer} = \text{US and cty. worker} \neq \text{US}}{\text{all jobs in } j \text{ where cty. employer} = \text{US}}. \tag{A.3.1}$$

Appendix Table A3 reports this measure and shows that it is very similar to that in equation (1).

A.3.2 Offshoring across categories in the SOC system

To make our measure easier to use in future research, we compute the fraction of jobs offshored for the SOC categories represented in our data. Figure A.9 plots the measure in (1) when computed for the categories on the platform (y-axis) vs. the SOC categories (x-axis). The categories on the platform are often more disaggregated than those in the SOC, so the figure often contains many occupations on the y-axis corresponding to a single point on the x-axis. The figure shows that, while the measures are positively correlated, the SOC categories are often too broad and mask substantial heterogeneity in the extent to which different occupations are being offshored. For example, the SOC category "Search Marketing Strategists" includes a wide range of more specific occupations on the platform. Within this SOC category, we observe a difference of 29 percentage points in the probability of offshoring jobs between "E-commerce Programmers and Developers" and "Display Advertising Specialists" ($\mathcal{O}_j = 0.79$ and $\mathcal{O}_j = 0.50$, respectively). This also suggests that having more disaggregated job categories than those currently available in official statistics can better capture the degree to which different jobs are offshored and other important dimensions of international labor transactions.

Figure A.9: Offshoring within SOC categories



Notes: Each circle represents an occupation. The figure compares the frequency with which jobs are offshored using equation (1) for SOC categories vs. platform categories.

A.4 Additional empirical exercises

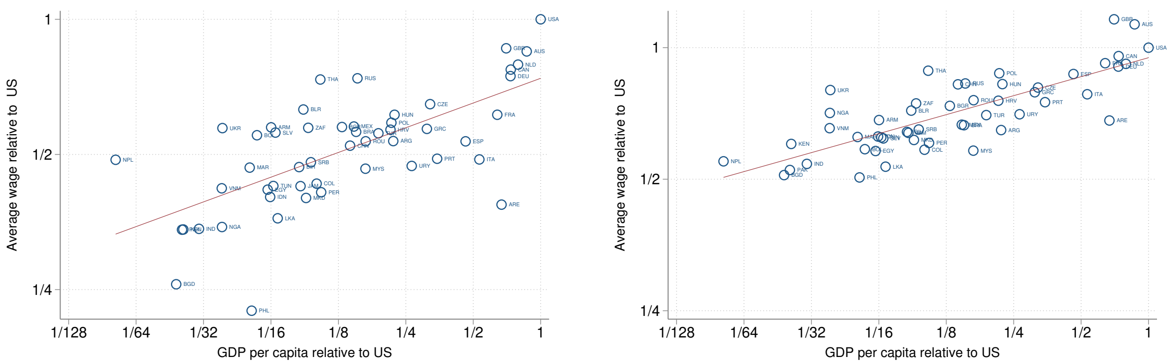
This section provides additional details about the auxiliary exercises reported in Section 3.2.

A.4.1 External validity

To assess whether the cross-country wage gradient is a feature of remote labor markets more broadly, or specific to our platform, we collected data from one of the largest competing platforms. These data include posted (ask) wages and worker locations for approximately 151,000 workers. We restrict attention to the set of countries that appear in both platforms; relative to our baseline sample, this drops Bermuda because data are not available on the competing platform. For comparability, we estimate the gradient on each platform using *ask* wages (since the competing platform does not report transacted wages) and without the residualization applied in equation (2) (since we do not have data on controls for the alternative platform). Appendix Figure A.10 plots the resulting average wages against log GDP per capita for both platforms. The estimated slope is 0.18 (SE 0.03) on our platform and 0.14 (SE 0.02) on the competing platform. Two observations are worth noting. First, the slope on our platform (0.18) is somewhat smaller than the headline estimate of 0.22 reported in Figure 4a, reflecting the use of nonresidual ask wages and the exclusion of Bermuda. Second, and most importantly, the gradients are quantitatively similar across platforms, suggesting that the cross-country compression of remote wages documented in this paper is a feature of online remote labor markets rather than an artifact of selection into the particular platform we study.

Figure A.10: Cross-country wage gradients on our platform versus an alternative platform

(a) Our platform (b) Alternative platform



Notes: The x-axes report the log of relative GDP per capita in US dollars (World Development Indicators). The y-axes plot average country-level ask wages relative to the US. The outcome in panel (a) is ask wages from our platform; in panel (b), ask wages from an alternative remote-work platform. Both panels include the same sample of countries. Relative to our baseline sample of countries, this sample excludes Bermuda because data are not available on the alternative platform. The red lines show the linear fit of the data. The estimated slope in the left panel is 0.18 (0.03) and in the right panel is 0.14 (0.02).

A.4.2 Disentangling sources of cross-country wage differences

Controlling for employer fixed effects: We estimate a specification that uses unique employer identifiers to control for employer fixed effects in equation (2). We estimate this regression using the sample of employers for which we observe more than one worker; this subsample accounts for 42% of the observations (unfortunately, we do not observe all the workers hired by each employer). Appendix Figure A.4, panel (a), plots the resulting average wages in each location after residualizing with these unique employer fixed effects. As noted in the paper, the figure continues to show a strong relationship between the (residual) remote wages and the GDP per capita of the workers' location, although the slope of this relationship drops to 0.15 (SE 0.02). This shows that even when working for the same employer, remote workers from richer countries earn higher wages.

Controlling for worker fixed effects (pricing to market): We estimate a version of (2) that substitutes worker-level controls with unique worker fixed effects. Appendix Figure A.4, panel (b), plots the wages paid by employers from each country, obtained from the employer-country dummies $\mathbb{D}_f$ in this regression, for the set of countries that have more than 100 workers. Workers are paid somewhat more when working for employers from richer countries, although the relationship is mild and driven by only a few countries (slope of 0.05 with a standard error of 0.02).

Trade costs: Appendix Figure A.5 plots average wages across workers' and employers' locations, obtained from a version of (2) that incorporates controls for the time difference and geographical distance between the employer's and the worker's countries, and for whether the countries share a common language, currency, and legal origin. The figure shows that adding these controls does not change the main results shown in Figures 4a and 4b.

Border effects: We compare average wages across US states after residualizing them for worker characteristics. Unfortunately, we do not observe the transacted wage for a sufficient number of workers and employers in each US state to estimate (2) at the state level (there are only 12 states with more than 100 workers that report these data). Thus, we use data on ask wages for workers located in the US to estimate:

$$w_i = \mathbb{S}_i + \boldsymbol{\beta}'\mathbf{X}_i + \varepsilon_i. \tag{A.4.1}$$

Here, w_i is the ask wage of worker i, and $\mathbb{S}_i$ denotes a full set of state fixed effects for the worker's state. The omitted state is California—the state with the most workers in our sample—so the state fixed effects measure average wages in each state relative to the average wage earned in California. Since equation (A.4.1) is estimated on the ask wage data, we cannot control for the location of the employer (workers only post one ask wage in their profiles). We also exclude states with fewer than 30 workers (North Dakota, Wyoming, and Alaska, with 18, 25, and 26 workers, respectively).

Appendix Figure A.6 compares relative wages to the relative GDP per capita of each of the 47 states with at least 30 workers in our sample. It shows that the pattern across US states is similar to the one we observe across countries: Workers from richer states earn higher wages on average. The slope of this relationship is 0.26 (SE 0.04), and the R^2 is 0.48. These patterns are remarkably similar to the cross-country patterns documented above and suggest that the worker's location plays a large role in shaping wages, but not predominantly because of border effects.

Comparison with non-remote wages and local prices: We obtain data on non-remote wages for occupations that are similar to those represented on the platform from the International Comparison Program (ICP) from the World Bank. We include the following occupations in the ICP database: Accounting and Bookkeeping Clerks, HR Professionals, Computer Operators, Data Processing Managers, and Database Administrators. The first panel in Appendix Figure A.7 shows that the relationship between remote wages and non-remote wages from similar occupations resembles that in Figure 4a. The second panel compares remote wages to local price levels and shows that remote wages are higher for workers living in countries with higher price levels.

A.5 Derivation of Equations (20) and (21)

Log differentiating (18), we can write the change in worker i's wage as

$$dw_{ij} = \frac{\theta}{\rho+\theta} db_{jc} + \frac{\rho-\epsilon}{\rho+\theta} d\omega_j + \frac{\epsilon-1}{\rho+\theta} dp_j + d\varphi_{jc} + dz_{ij}.$$

The average change in remote wages in sector j is:

$$dw_j \equiv \sum_c s_{jc} \mathbb{E}_c \left[dw_{ij} \right] = \sum_c s_{jc} \mathbb{E}_c \left[d\omega_{jc} \right] + \sum_c s_{jc} dz_{jc} = d\omega_j + da_j + dz_j,$$

where $dx_j \equiv \sum_c s_{jc} dx_{jc}$ denotes the sector-j cross-country average change in a variable x. Combining these equations with (19), we obtain:

$$dw_{ij} = \frac{\theta}{\rho+\theta} de_c + \frac{\rho-\epsilon}{\rho+\theta} dw_j + \frac{\epsilon-1}{\rho+\theta} dp_j + d\psi_{jc} + dz_{ij},$$

where $d\psi_{jc} \equiv d\varphi_{jc} - \frac{\rho-\epsilon}{\rho+\theta}\left[dz_j + da_j \right] + \frac{\theta}{\rho+\theta}\left[\pi_c + g_{jc} \right]$. The average change in wages is

$$dw_j = \frac{\theta}{\theta+\epsilon} de_j + \frac{\epsilon-1}{\theta+\epsilon} dp_j + d\phi_j,$$

where $d\phi_j \equiv \frac{\rho+\theta}{\theta+\epsilon}\left[d\psi_j + dz_j \right]$, which corresponds to equation (21) in the text.

A.6 Model extensions

This Appendix describes three extensions to our baseline model in Section 4. First, we allow for a more general utility function under which expenditures on sectoral goods are functions of prices. Second, we provide an alternative formulation for the demand for remote work in which workers from different countries are perfect substitutes but specialize in the production of different tasks.

A.6.1 Allowing for changes in total expenditures on sectoral goods

The utility function presented in equation (7) implies that dollar expenditures on each sector are fixed (equation (8)). This section presents a more general preference structure that relaxes this result. In particular, we assume that the utility for the representative US household is given by:

$$U(M, Y) = M + \frac{1}{\zeta} Y^{\zeta},$$

with $\zeta < 1$, where

$$Y = \left[\sum_j \bar{\gamma}_j^{\frac{1}{\eta}} Y_j^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}},$$

is a CES aggregate over the sectoral goods Y_j. Cost minimization implies that the demand for good j is given by

$$P_j Y_j = \bar{\gamma}_j \left[\frac{P_j}{P} \right]^{1-\eta} P^{\frac{\zeta}{\zeta-1}}, \tag{A.6.1}$$

where $P \equiv \left[\sum_j \bar{\gamma}_j P_j^{1-\eta} \right]^{\frac{1}{1-\eta}}$ is the aggregate price index. Remote labor demand from country c in sector j is

$$\Omega_{jc} N_{jc} = \bar{\gamma}_j \bar{\alpha}_j A_{jc}^{\rho-1} \left[\frac{\Omega_{jc}}{\Omega_j} \right]^{1-\rho} \left[\frac{\Omega_j}{P_j} \right]^{1-\epsilon} \left[\frac{P_j}{P} \right]^{1-\eta} P^{\frac{\zeta}{\zeta-1}}. \tag{A.6.2}$$

Note that these preferences nest the equations in the text for the special case of $\eta = 1$ and $\zeta \to 0$. We evaluate the sensitivity of our results to alternative values of η and ζ in Appendix Table A13. The second panel of the table shows results for a higher elasticity of substitution across sectors, $\eta = 2$. When sectoral goods are substitutes, the increase in their price triggered by reshoring remote jobs shifts expenditure from the most exposed sectors (Design and Web) towards the least exposed sectors (Admin and Writing). As a consequence, remote and non-remote US wages in Admin and Writing rise more in this calibration relative to the baseline, while wage increases in Design and Web are more muted. The magnitude of the wage and employment changes is, however, very close to that in the baseline calibration.

A.6.2 Alternative occupation production function

We present an alternative version of the model in Section 4.1, in which workers from different countries are perfect substitutes, but specialize in different tasks. We demonstrate that this version is isomorphic to the more parsimonious model analyzed in the main text. Specifically, we modify the framework from Section 4.1 by assuming that remote output in sector j is produced by a continuum of tasks indexed by $\iota \in [0,1]$:

$$N_j = \left[\int_0^1 y_j(\iota)^{\frac{\sigma_j - 1}{\sigma_j}} d\iota \right]^{\frac{\sigma_j}{\sigma_j - 1}}, \tag{A.6.3}$$

where $y_j(\iota)$ denotes the quantity of task ι used as input in sector j, and σ_j is the elasticity of substitution across tasks within sector j. Each task ι can be produced remotely by workers in different countries c. The cost of purchasing task ι from country c is $\Omega_{jc}/x_{jc}(\iota)$, where Ω_{jc} is the wage per efficient unit of labor from country c and sector j, and $x_{jc}(\iota)^{-1}$ is the number of efficiency units of labor from country c required to produce task ι. These requirements can be country-task specific, indicating that labor from different countries can be relatively more productive in different tasks. Alternatively, x_{jc} can be interpreted as reflecting contracting costs that lead to different efficiency units of labor required to complete the task.

We assume that efficiency units of labor from different countries are perfect substitutes in the production of a task, so tasks are supplied by the lowest cost location. Consequently, the price paid on the platform for task ι in sector j is $p_j(\iota) = \min \left\{ \frac{\Omega_{j1}}{x_{j1}(\iota)}, \frac{\Omega_{j2}}{x_{j2}(\iota)}, ... \right\}$. Furthermore, we assume that $x_{jc}(\iota)$ is a random variable drawn independently for each task ι from a Fréchet distribution given by

$$G_{jc}(x) \equiv \Pr\left(x_{jc}(\iota) \leq x\right) = e^{-\tilde{A}_{jc} x^{1-\rho}},$$

with shape parameter $\rho > 2$, and scale parameter $\tilde{A}_{jc} > 0$. A lower value of ρ implies that the draws $x_{jc}(\iota)$ are more dispersed across tasks, so that differences in comparative advantage across tasks are stronger. A larger value of $\tilde{A}_{jc}$ implies that workers from a country are likely to be more productive across all tasks, representing an absolute advantage.

The distributional assumption implies that the distribution of prices on the platform for task ι, $p_j(\iota)$, is also Fréchet. This distribution, denoted by $\tilde{G}_j(p)$, is given by

$$\tilde{G}_j(p) = 1 - \prod_c \Pr\left(\frac{\Omega_{jc}}{x_{jc}(\iota)} > p\right) = 1 - e^{-\Phi_j p^{\rho-1}},$$

with $\Phi_j \equiv \sum_c \tilde{A}_{jc}\Omega_{jc}^{1-\rho}$.

We can now compute the cost function associated with (A.6.3). The cost function of sector j is a weighted average of the tasks' prices given by

$$\Omega_j = \Delta_j \Phi_j^{\frac{1}{1-\rho}}, \tag{A.6.4}$$

where $\Delta_j \equiv \Gamma\left(\frac{\rho-\sigma_j}{\rho-1}\right)^{\frac{1}{1-\sigma_j}}$, and $\Gamma(\cdot)$ denotes the Gamma function. We assume that $\sigma_j < \rho$ so that the integral defining $\Omega_j^{1-\sigma_j}$ converges.[31]

The probability that a task with a labor requirement $x_{jc}(\iota)$ is supplied by country c in sector j is

$$\Pr\left(\frac{\Omega_{jc}}{x_{jc}(\iota)} \leq \min_{s\neq c}\left\{\frac{\Omega_{js}}{x_{js}(\iota)}\right\}\right),$$

which is equal to

$$\begin{aligned}\prod_{s\neq c}\Pr\left(\frac{\Omega_{js}}{x_{js}(\iota)} \geq \frac{\Omega_{jc}}{x_{jc}(\iota)}\right) &= \prod_{s\neq c} e^{-\tilde{A}_{js}\left[\frac{\Omega_{js}}{\Omega_{jc}}x_{jc}(\iota)\right]^{1-\rho}} \\ &= e^{\left[x_{jc}(\iota)\right]^{1-\rho}\left[\tilde{A}_{jc}-\Phi_j\Omega_{jc}^{\rho-1}\right]}.\end{aligned}$$

Integrating across all possible values of $x_{jc}(\iota)$, we obtain the probability that country c

[31] Given that the production function of sector j combines tasks with a CES technology, the cost function is given by:

$$\Omega_j^{1-\sigma_j} = \int_0^1 p_j(\iota)^{1-\sigma_j}\, d\iota.$$

The moment generating function for $y = -\ln(p)$ is $\mathbb{E}\left(e^{ty}\right) = \Gamma\left(1-\frac{t}{\rho-1}\right)\Phi_j^{\frac{t}{\rho-1}}$. Then, $\mathbb{E}\left(e^{yt}\right)^{-1/t} = \Gamma\left(1-\frac{t}{\rho-1}\right)^{-1/t}\Phi_j^{\frac{-1}{\rho-1}}$. The expression for the cost function follows by replacing t with $\sigma_j - 1$ (see Eaton and Kortum, 2002).

supplies the task:[32]

$$s_{jc} = \frac{\tilde{A}_{jc}\Omega_{jc}^{1-\rho}}{\Phi_j}.$$

Under our distributional assumptions, the probability that a location supplies an individual task coincides with the share of spending on tasks performed in the location (see Eaton and Kortum, 2002). That is,

$$\frac{\tilde{A}_{jc}\Omega_{jc}^{1-\rho}}{\Phi_j} = s_{jc} = \frac{\Omega_{jc}N_{jc}}{\Omega_j N_j}.$$

Substituting (A.6.4), we obtain the demand for efficiency units of labor from location c in sector j:

$$\Omega_{jc}N_{jc} = \tilde{A}_{jc}\Delta_j^{\rho-1}\left[\frac{\Omega_{jc}}{\Omega_j}\right]^{1-\rho}\Omega_j N_j,$$

which, in combination with the demand for remote output $\Omega_j N_j = \bar{\alpha}_j \left[\frac{\Omega_j}{P_j}\right]^{1-\epsilon}\bar{\gamma}_j$, yields equation (10) with $A_{jc} = \tilde{A}_{jc}^{\frac{1}{\rho-1}}\Delta_j$.

A.7 Model results

[32] This integral is given by

$$\begin{aligned} s_{jc} &= \int_0^\infty e^{x^{1-\rho}\left[\tilde{A}_{jc}-\Phi_j[\Omega_{jc}]^{\rho-1}\right]}\tilde{A}_{jc}x^{-\rho}\left[\rho-1\right]e^{-\tilde{A}_{jc}x^{1-\rho}}dx \\ &= \int_0^\infty e^{-x^{1-\rho}\Phi_j[\Omega_{jc}]^{\rho-1}}\tilde{A}_{jc}x^{-\rho}\left[\rho-1\right]dx \\ &= \tilde{A}_{jc}\left[\rho-1\right]\int_0^\infty x^{-\rho}e^{-x^{1-\rho}\Phi_j[\Omega_{jc}]^{\rho-1}}dx. \end{aligned}$$

Define $y \equiv \left[\Omega_{jc}\right]^{\rho-1}\Phi_j x^{1-\rho}$. Then, $dy = -\left[\Omega_{jc}\right]^{\rho-1}\Phi_j\left[\rho-1\right]x^{-\rho}dx$. This implies that the previous expression can be rewritten as follows:

$$s_{jc} = \frac{\tilde{A}_{jc}}{\left[\Omega_{jc}\right]^{\rho-1}\Phi_j}\int_0^\infty e^{-y}dy = \frac{\tilde{A}_{jc}\left[\Omega_{jc}\right]^{1-\rho}}{\Phi_j}.$$

Table A12: Reshoring Remote Jobs under Alternative Calibrations

	Sector			
Calibration	Admin	Design	Web	Writing
i. Baseline				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	12.9	22.5	22.3	9.2
Non-Remote US wage: $B_{jU} - 1$ (%)	1.8	10.3	7.7	1.7
Remote US labor: $\hat{N}_{jU} - 1$ (%)	100.2	101.1	133.7	61.0
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-0.9	-5.1	-4.2	-0.8
ii. ϵ half (recalibrated)				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	21.4	30.8	33.5	14.6
Non-Remote US wage: $B_{jU} - 1$ (%)	1.6	9.5	7.5	1.5
Remote US labor: $\hat{N}_{jU} - 1$ (%)	126.4	126.4	170.4	75.2
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-1.1	-6.4	-5.3	-1.0
iii. ϵ double (recalibrated)				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	8.1	17.6	15.8	6.1
Non-Remote US wage: $B_{jU} - 1$ (%)	1.9	10.7	7.6	1.8
Remote US labor: $\hat{N}_{jU} - 1$ (%)	71.3	72.7	93.9	44.7
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-0.6	-3.7	-2.9	-0.6
iv. θ half				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	18.6	28.5	29.4	13.0
Non-Remote US wage: $B_{jU} - 1$ (%)	1.7	9.8	7.0	1.6
Remote US labor: $\hat{N}_{jU} - 1$ (%)	67.5	69.2	88.9	42.5
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-0.6	-3.5	-2.8	-0.6
v. ρ double				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	13.6	23.2	23.4	9.6
Non-Remote US wage: $B_{jU} - 1$ (%)	1.8	10.3	7.8	1.6
Remote US labor: $\hat{N}_{jU} - 1$ (%)	109.2	110.0	146.4	65.9
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-1.0	-5.5	-4.6	-0.9

Notes: Panel (i) corresponds to Panel (i) in Table 5. Panels (ii) and (iii) vary the value of ϵ and recalibrate θ and ρ as described in Section 5.5. The parameter values in Panel (ii) are $\theta = 4.6$ and $\rho = 16.9$, and in Panel (iii) they are $\theta = 9$ and $\rho = 33.1$. Panels (iv) and (v) vary one parameter at a time.

Table A13: Reshoring Remote Jobs under Extended Model

Calibration	Sector			
	Admin	Design	Web	Writing
i. Baseline				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	12.9	22.5	22.3	9.2
Non-Remote US wage: $\hat{B}_{jU} - 1$ (%)	1.8	10.3	7.7	1.7
Remote US labor: $\hat{N}_{jU} - 1$ (%)	100.2	101.1	133.7	61.0
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-0.9	-5.1	-4.2	-0.8
ii. $\eta = 2$				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	13.9	17.8	19.4	10.2
Non-Remote US wage: $\hat{B}_{jU} - 1$ (%)	2.7	6.2	5.2	2.7
Remote US labor: $\hat{N}_{jU} - 1$ (%)	100.2	101.1	133.7	61.0
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-0.9	-5.1	-4.2	-0.8
iii. $\zeta = 0.5$				
Remote US wage: $\hat{\Omega}_{jU} - 1$ (%)	10.7	20.1	19.9	7.0
Non-Remote US wage: $\hat{B}_{jU} - 1$ (%)	-0.2	8.2	5.6	-0.3
Remote US labor: $\hat{N}_{jU} - 1$ (%)	100.2	101.1	133.7	61.0
Non-Remote US labor: $\hat{L}_{jU} - 1$ (%)	-0.9	-5.1	-4.2	-0.8

Notes: Panel (i) corresponds to Panel (i) in Table 5. Panels (ii) and (iii) correspond to the extended version of the model described in Appendix A.6.1. The parameter values for η and ζ in each case are as follows: (i) $\eta = 1$ and $\zeta \to 0$, (ii) $\eta = 2$ and $\zeta \to 0$, (iii) $\eta = 1$ and $\zeta = 0.5$.